

Biochemistry II

Carbohydrate and lipid metabolism, nitrogen pathways and integrated clinical reasoning

Updated 23 September 2026

Lecture- and book-grounded metabolism review expanded against 641 source-question occurrences in 14 supplied past-paper collections and one separately labelled reconstruction. Core pathways are retained; new material covers diagnostic enzymes, liver and renal biochemistry, endocrine signaling and detailed metabolic disorders. The original carbohydrate and first lipid slide decks were located in the wider BIOCHEM folder during this audit. Clinical extensions are distinguished from confirmed lecture content. This is coverage of the available material, not an official examination blueprint or a guarantee of future exam coverage.

How to use this book

Lecture-grounded material and book extensions are distinguished. Green/blue panels consolidate the key ideas; amber panels flag missing evidence; orange panels correct traps. This is a detailed review of the available sources, not an official exam blueprint. Original labeled figures remain important. Numerical laboratory ranges and procedures depend on the course method and local SOP.

Original lecture transcripts are preserved as PDF attachments and in the editable source folder when available. Automated transcripts can contain errors; study the checked synthesis first. Personal study only: original book/lecture illustrations retain their owners' rights.

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Reading map and source confidence

SOURCE BASIS: lecture + book

Use the lecture-supported carbohydrate and dietary-lipid sections first. Original slide PDFs, reconstructed notes and recordings are now cross-referenced. Book/reference extensions supply clinical comparisons tested in the supplied papers. Their inclusion establishes coverage of those questions, not of an unseen official syllabus. Raw transcripts are preserved, with transcription errors corrected in the synthesis.

Topic	Local evidence	Local Lippincott PDF reading
Glycolysis, other sugars, glycogen, gluconeogenesis, TCA	103-page original carbohydrate PDF, two note sets and three carbohydrate recordings	Chs. 8-12, pp. 176-276
Bioenergetics and pentose phosphate pathway	Mentioned or required to interpret carbohydrate lectures; expanded from book	Ch. 6, pp. 135-160; Ch. 13, pp. 277-297
Dietary lipids, absorption and chylomicrons	42-page original lipid PDF and three parts of Esmaeili lipid lecture 1	Ch. 15, pp. 326-339; relevant Ch. 18, pp. 420-432
Fatty acids, ketones, complex lipids and cholesterol	Book extension beyond the available first lipid lecture	Chs. 16-18, pp. 340-447
Nitrogen, amino acids and specialized products	Book extension; no corresponding course lecture found	Chs. 19-21, pp. 448-531
Nucleotides and integrated metabolism	Book extension; confirmed question themes in supplied past papers	Chs. 22-26, pp. 532-650
Nutrition and vitamin mechanisms	Targeted book extension, not the full nutrition syllabus	Chs. 27-28, pp. 651-720

Corrections to remember

Glycolysis itself does not consume oxygen. Each pyruvate has three carbons. Fructose 2,6-bisphosphate activates PFK-1, whereas fructose 1,6-bisphosphate feeds forward to pyruvate kinase. The urea cycle uses three ATP molecules but four high-energy phosphate bonds. Acetyl-CoA from even-chain fatty acids cannot provide net glucose in humans.

September 2026 coverage update

The new audit adds the previously absent clinical-diagnostic and endocrine topics and expands fine biochemical mechanisms. The row-level audit records a review destination for every source occurrence, including a separate warning for each of the 23 ungraded items. A destination means the underlying concept is taught; it does not authenticate the source answer key.

Newly found lecture evidence: carbohydrate metabolism MD Kish.pdf, 103 pages; Lipid metabolism 1.pdf, 42 pages; and the 41-page enzyme kinetics/regulation PDF. The first two agree with the available metabolism recordings and supply the original diagrams. The enzyme deck is supporting course material, not proof that every clinical enzyme test was lectured in Term 2.

Sources for this section

Biochemistry Metabolism (Part 1) Notes - Hawraa.pdf - PDF pp. 1-7; all handwritten pages visually inspected

Biochemistry metabolism notes (pt 2) slides 27-80 .pdf - PDF pp. 1-6

IMG_4406.transcript.txt - entire transcript

IMG_4407.transcript.txt - entire transcript

Biochemistry Metabolism Recorded Lecture - 1h04m20s.transcript.txt - entire transcript

Lipid metabolism Esmaeili 1.1.transcript.txt - entire transcript

Lipid metabolism Esmaeili 1.2.transcript.txt - entire transcript

Lipid metabolism1 Esmaeili 1.3.transcript.txt - entire transcript

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - PDF p. 4 (edition); contents pp. 8-9

carbohydrate metabolism MD Kish.pdf - PDF pp. 1-103; pathway and regulation cross-check

Lipid metabolism 1.pdf - PDF pp. 1-42; lipid digestion, chylomicrons and disorders

Compartmentation, energy and pathway regulation

SOURCE BASIS: lecture + book

Metabolism is an interconnected network. Catabolic pathways release usable energy and reducing equivalents while breaking molecules into smaller products. Anabolic pathways use ATP and reducing power to build molecules. A pathway may be amphibolic: the TCA cycle oxidizes acetyl units but also provides intermediates for biosynthesis. Flux depends on substrate supply, enzyme activity, compartment access and demand for products. A reaction can be thermodynamically favorable yet slow if its activation barrier is high; enzymes lower the barrier without changing the reaction equilibrium.

Location	Major processes	Consequence
Cytosol	Glycolysis, pentose phosphate pathway, glycogen metabolism, fatty-acid synthesis	NADPH made in the cytosol can support reductive synthesis and antioxidant defense.
Mitochondrial matrix	PDH, most TCA enzymes, beta-oxidation, ketogenesis, pyruvate carboxylase, first two urea reactions	Mature RBCs cannot perform these pathways.
Inner mitochondrial membrane	Respiratory chain, ATP synthase, succinate dehydrogenase, specific carriers	An intact membrane and proton gradient couple oxidation to ATP synthesis.
ER	Glucose-6-phosphatase system, cholesterol regulation, fatty-acid elongation/desaturation, lipid assembly	Metabolite transport into or across this compartment can be a separate control point.
Lysosomes/peroxisomes	Macromolecule degradation / very-long-chain and branched-chain fatty-acid processing	A compartment defect can block a pathway even when the catalytic enzyme is otherwise present.

- High ATP and NADH generally signal abundant energy; AMP/ADP and NAD⁺ favor fuel oxidation. These are tendencies applied to particular enzymes, not universal on/off rules.

- NADH mainly conveys electrons toward energy production. NADPH mainly supports reductive biosynthesis, glutathione recycling, P450 reactions and the phagocyte oxidative burst. They are distinct coenzyme pools.
- Allosteric regulation can act rapidly. Reversible phosphorylation couples hormone signals to enzyme activity. Changes in transcription or protein degradation alter enzyme amount more slowly.
- Insulin binds a receptor tyrosine kinase. Hepatic glucagon signaling uses a G-protein-coupled receptor, adenylyl cyclase, cAMP and protein kinase A. Phosphorylation activates some enzymes and inhibits others.

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Biochemistry Metabolism (Part 1) Notes - Hawraa.pdf - PDF pp. 1-7; all handwritten pages visually inspected

Biochemistry metabolism notes (pt 2) slides 27-80 .pdf - PDF pp. 1-6

IMG_4406.transcript.txt - entire transcript

IMG_4407.transcript.txt - entire transcript

Biochemistry Metabolism Recorded Lecture - 1h04m20s.transcript.txt - entire transcript

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch. 6 pp. 135-158; Ch. 8 pp. 176-184; Ch. 24 pp. 584-608

Glucose transport, trapping and tissue priorities

SOURCE BASIS: lecture + book

Transport and phosphorylation are separate steps. Facilitated GLUT transport moves glucose down its concentration gradient. Phosphorylation traps the sugar as glucose-6-phosphate and keeps free intracellular glucose low. In the intestine and kidney, sodium-coupled glucose transport can move glucose against its gradient using the sodium gradient maintained by Na⁺/K⁺-ATPase.

Protein	Main teaching sites	Important distinction
GLUT1	RBCs, blood-brain barrier and widespread basal uptake	Insulin-independent basal transport.
GLUT3	Neurons	High-affinity uptake supports neuronal glucose use.
GLUT2	Liver, basolateral intestine/kidney; classic book pancreatic model	High capacity, bidirectional movement as gradients change. Hepatic uptake is not GLUT4-mediated.
GLUT4	Skeletal/cardiac muscle and adipose tissue	Insulin recruits transporters to the surface. Muscle contraction also promotes uptake.
GLUT5	Intestinal apical fructose transport	Fructose transport is not glucose transport.

Property	Hexokinase	Glucokinase
Affinity/capacity	Low Km, lower Vmax: functions at low glucose	Higher Km and Vmax: handles a carbohydrate load

Property	Hexokinase	Glucokinase
Product inhibition	Inhibited by glucose-6-phosphate	Not directly inhibited by glucose-6-phosphate
Tissue purpose	Protects basal glucose use in many tissues	Hepatic glucose handling and pancreatic glucose sensing
Regulation	Local product/energy demand	Hepatic expression increased by insulin; glucokinase regulatory protein responds to sugar phosphates

Tissue use is the exam anchor

RBCs always rely on glycolysis for ATP. The brain ordinarily depends heavily on glucose but adapts to use ketones during prolonged fasting; this is not an instantaneous guarantee of protection in acute hypoglycemia. Liver maintains blood glucose. Muscle retains its glycogen for its own work.

Hepatic GKR binds and sequesters glucokinase in the nucleus. Fructose-6-phosphate favors formation of the inactive glucokinase-GKR complex, whereas fructose-1-phosphate opposes it. Increased glucose also favors release and cytosolic activity. This regulation is not direct feedback inhibition of glucokinase by glucose-6-phosphate. Glucokinase inactivation can cause MODY2 by shifting glucose sensing and insulin secretion thresholds.

At the intestinal brush border, sucrase releases glucose plus fructose, lactase releases glucose plus galactose, and maltase releases glucose. SGLT1 handles apical glucose/galactose, GLUT5 apical fructose, and GLUT2 basolateral exit. Renal glucose reabsorption uses sodium-coupled transport, not insulin-controlled GLUT4. In diabetes the filtered load can exceed the transport maximum and cause glycosuria: insulin independence does not mean renal glucose handling is clinically unchanged.

Sources for this section

Biochemistry Metabolism (Part 1) Notes - Hawraa.pdf - PDF pp. 1-7; all handwritten pages visually inspected

Biochemistry metabolism notes (pt 2) slides 27-80 .pdf - PDF pp. 1-6

IMG_4406.transcript.txt - entire transcript

IMG_4407.transcript.txt - entire transcript

Biochemistry Metabolism Recorded Lecture - 1h04m20s.transcript.txt - entire transcript

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch. 8 pp. 185-190; summary pp. 202-203

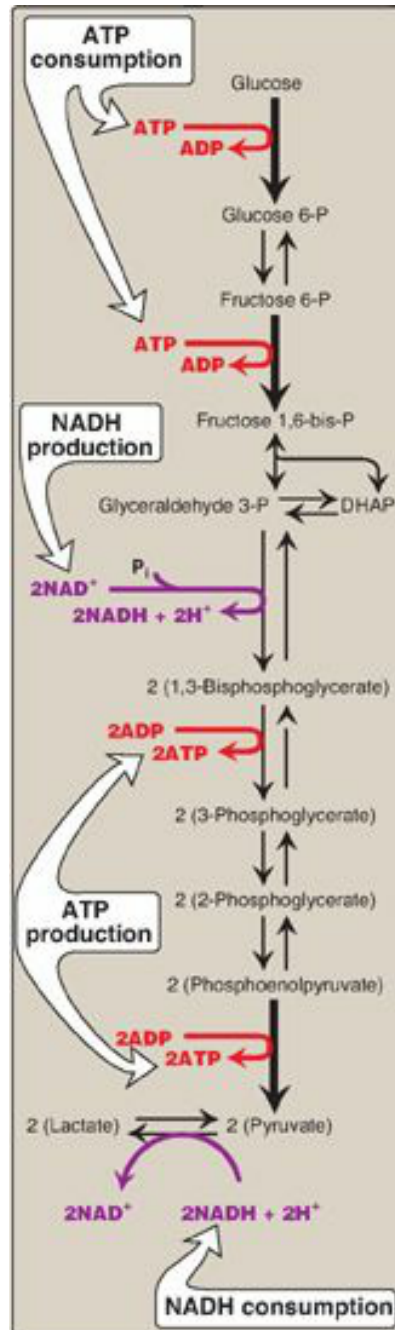
Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition - PDF pp. 176-187

Glycolysis: every step, carbon count and energy event

SOURCE BASIS: lecture + book

One six-carbon glucose forms two three-carbon pyruvates in the cytosol. The first phase spends two ATP. Cleavage and isomerization create two glyceraldehyde-3-phosphates, so every subsequent reaction occurs twice per starting glucose. The payoff phase produces four ATP and two NADH, giving a net two ATP. Molecular oxygen is not a reactant in any glycolytic step. Under aerobic conditions the reducing equivalents can be reoxidized through mitochondrial shuttles and the respiratory chain; under oxygen limitation or in RBCs, lactate formation regenerates cytosolic NAD⁺.

Step and enzyme	Conversion	Energy or key property
1 Hexokinase/glucokinase	Glucose to glucose-6-phosphate	Consumes ATP; physiologically irreversible.
2 Phosphoglucose isomerase	Glucose-6-phosphate to fructose-6-phosphate	Aldose/ketose interconversion.
3 PFK-1	Fructose-6-phosphate to fructose-1,6-bisphosphate	Consumes ATP; committed, regulated, irreversible step.
4 Aldolase	Fructose-1,6-bisphosphate to GAP plus DHAP	Six carbons become two three-carbon compounds.
5 Triose phosphate isomerase	DHAP to GAP	Two GAP molecules proceed through payoff.
6 GAP dehydrogenase	GAP plus Pi to 1,3-bisphosphoglycerate	NAD ⁺ becomes NADH; no ATP supplies this added phosphate.
7 Phosphoglycerate kinase	1,3-BPG to 3-phosphoglycerate	Substrate-level phosphorylation produces ATP.
8 Phosphoglycerate mutase	3-phosphoglycerate to 2-phosphoglycerate	Moves the existing phosphate within the molecule.
9 Enolase	2-phosphoglycerate to phosphoenolpyruvate	Dehydration creates high transfer-potential PEP.
10 Pyruvate kinase	PEP to pyruvate	Substrate-level phosphorylation; irreversible.



Source pathway: trace the two ATP investments, two doubled ATP-producing steps and regeneration of NAD⁺ during lactate formation. The diagram is a compact source image; use the step table for enzyme names.

Identify: Glucose-6-phosphate; Fructose-1,6-bisphosphate; DHAP/GAP; 1,3-BPG; PEP; Pyruvate; Lactate

Figure source: Lippincott Biochemistry 6e, PDF p. 198, extracted source diagram

Check yourself: If GAP dehydrogenase lacks NAD⁺, can abundant glucose and ADP keep glycolysis running?

No. Oxidation of GAP cannot proceed, so the downstream ATP-producing steps are starved of intermediates. Lactate dehydrogenase normally recycles NADH to NAD⁺ under anaerobic conditions. ATP availability alone does not replace the required electron acceptor.

Sources for this section

Biochemistry Metabolism (Part 1) Notes - Hawraa.pdf - PDF pp. 1-7; all handwritten pages visually inspected

Biochemistry metabolism notes (pt 2) slides 27-80 .pdf - PDF pp. 1-6

IMG_4406.transcript.txt - entire transcript

IMG_4407.transcript.txt - entire transcript

Biochemistry Metabolism Recorded Lecture - 1h04m20s.transcript.txt - entire transcript

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch. 8 pp. 188-204

PFK-1, pyruvate kinase and the lactate shunt

SOURCE BASIS: lecture + book

Control	Activation	Inhibition and implication
PFK-1	AMP and fructose-2,6-bisphosphate	ATP and citrate suppress flux when energy and biosynthetic carbon are abundant.
Hepatic PFK-2/FBPase-2	Insulin favors dephosphorylation, raising fructose-2,6-bisphosphate	Glucagon/PKA phosphorylation lowers fructose-2,6-bisphosphate in the liver, opposing glycolysis.
Pyruvate kinase	Fructose-1,6-bisphosphate feed-forward activation	ATP and alanine inhibit; hepatic glucagon/PKA phosphorylation inhibits.
Hexokinase	Substrate availability	Glucose-6-phosphate feedback inhibition.

Lactate dehydrogenase catalyzes pyruvate + NADH + H⁺ to lactate + NAD⁺. This reversible reaction conserves the carbon skeleton while regenerating NAD⁺. It produces no additional ATP. In the Cori cycle, lactate travels to liver for gluconeogenesis, transferring part of the energetic burden to liver. Alcoholic fermentation makes acetaldehyde and then ethanol in microorganisms. Humans do not use pyruvate decarboxylase to ferment glucose to ethanol.

In RBCs, diversion of 1,3-BPG through the 2,3-BPG shunt changes hemoglobin oxygen affinity while bypassing an ATP-producing reaction. Pyruvate kinase deficiency reduces RBC ATP and can cause hemolysis. This differs from G6PD deficiency, where the primary failure is NADPH-dependent antioxidant protection, not a direct ATP-producing step.

Correcting the lecture transcription

The early transcript and handwritten p.6 describe PFK inhibition by fructose-1,6-bisphosphate. The later lecture correctly discusses ATP and citrate inhibition. The standard distinction is PFK-1 activation by fructose-2,6-bisphosphate versus pyruvate-kinase activation by fructose-1,6-bisphosphate. Do not memorize the early error.

The Rapoport-Luebering branch uses bisphosphoglycerate mutase to form 2,3-BPG from 1,3-BPG, then bisphosphoglycerate phosphatase to form 3-phosphoglycerate. It bypasses phosphoglycerate kinase and sacrifices one ATP per diverted triose. Phosphoglycerate kinase is reversible: it produces ATP in glycolysis and consumes ATP in gluconeogenesis.

Lactate is not simply a poison

Lactate production regenerates NAD⁺ and can continue even when oxygen is present. Exercise fatigue is multifactorial; lactate alone is not the cause of delayed muscle soreness. Some old MCQs use an oversimplified lactate/fatigue association. Fluoride inhibits enolase; arsenate uncouples ATP capture at the GAP-dehydrogenase/phosphoglycerate-kinase sequence, whereas arsenite principally inhibits lipoate-dependent complexes.

For the paired hemolysis/acidosis question, pyruvate-kinase deficiency lowers RBC ATP and causes hemolytic anemia. The mercury option invokes the traditional teaching model of sulfhydryl-enzyme interference with pyruvate oxidation. Mercury species and experimental conditions matter: methylmercury studies also show altered mitochondrial pyruvate transport, not uniformly strong PDH inhibition. Do not turn that exam association into a mercury-specific clinical diagnostic rule. Thiamine deficiency and inherited PDH defects are clearer mechanisms for pyruvate/lactate accumulation.

Sources for this section

Biochemistry Metabolism (Part 1) Notes - Hawraa.pdf - PDF pp. 1-7; all handwritten pages visually inspected

Biochemistry metabolism notes (pt 2) slides 27-80 .pdf - PDF pp. 1-6

IMG_4406.transcript.txt - entire transcript

IMG_4407.transcript.txt - entire transcript

Biochemistry Metabolism Recorded Lecture - 1h04m20s.transcript.txt - entire transcript

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch. 8 pp. 190-203

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition - PDF pp. 189-207

carbohydrate metabolism MD Kish.pdf - PDF pp. 8-26

Alexandria National University: Medical Biochemistry Unit, oxidative decarboxylation teaching note - Accessed 23 September 2026; supplemental clinical verification - <https://medportal.anu.edu.eg/moodledata/filedir/4e/92/4e929053b3f6ed5251a83e21ed7a3b38bd2c8612>

Transport of pyruvate into mitochondria is involved in methylmercury toxicity (cell study) - Accessed 23 September 2026; supplemental clinical verification - <https://www.nature.com/articles/srep21528>

PDH and the TCA cycle: enzymes, cofactors and regulation

SOURCE BASIS: lecture + book

Pyruvate enters mitochondria and undergoes irreversible oxidative decarboxylation by the pyruvate dehydrogenase complex: $\text{pyruvate} + \text{CoA} + \text{NAD}^+ \rightarrow \text{acetyl-CoA} + \text{CO}_2 + \text{NADH}$. The two-carbon acetyl group enters the TCA cycle by condensing with four-carbon oxaloacetate. PDH is a bridge reaction, not one of the eight cycle reactions. Its cofactors are TPP (B1), lipoate, CoA (B5), FAD (B2) and NAD^+ (B3). The alpha-ketoglutarate and branched-chain ketoacid dehydrogenase complexes use the same five-cofactor pattern.

TCA enzyme	Reaction	Output/control
Citrate synthase	Oxaloacetate + acetyl-CoA to citrate	Condensation; citrate can inhibit.
Aconitase	Citrate through cis-aconitate to isocitrate	Isomerization.
Isocitrate dehydrogenase	Isocitrate to alpha-ketoglutarate	$\text{CO}_2 + \text{NADH}$; $\text{ADP}/\text{Ca}^{2+}$ stimulate, ATP/NADH inhibit.
Alpha-ketoglutarate dehydrogenase	Alpha-ketoglutarate to succinyl-CoA	$\text{CO}_2 + \text{NADH}$; Ca^{2+} stimulates, $\text{NADH}/\text{succinyl-CoA}$ inhibit.
Succinyl-CoA synthetase	Succinyl-CoA to succinate	GTP or ATP depending on isoform; substrate-level phosphorylation.

TCA enzyme	Reaction	Output/control
Succinate dehydrogenase	Succinate to fumarate	FAD-linked oxidation; also respiratory complex II in inner membrane.
Fumarase	Fumarate to malate	Hydration.
Malate dehydrogenase	Malate to oxaloacetate	NADH; oxaloacetate consumption helps pull the reaction forward.

- Per acetyl-CoA, the cycle yields three NADH, one FADH₂, one GTP and two CO₂. The two newly entering acetyl carbons are not necessarily the two carbons lost as CO₂ on the first turn.
- PDH kinase phosphorylates and inhibits PDH. High ATP, NADH and acetyl-CoA favor this inhibition. PDH phosphatase removes the phosphate; calcium stimulates activity in working muscle.
- The cycle does not directly consume molecular oxygen, but depends on the respiratory chain to regenerate oxidized cofactors. Respiratory failure increases NADH and slows TCA flux.
- Pyruvate carboxylase replenishes oxaloacetate using biotin, ATP and bicarbonate. Acetyl-CoA activates it, coordinating incoming acetyl units with availability of the accepting four-carbon pool.
- Citrate can leave for fatty-acid synthesis; alpha-ketoglutarate and oxaloacetate support amino-acid synthesis; succinyl-CoA supports heme synthesis. Withdrawal must be balanced by anaplerosis.

Why PDH or thiamine deficiency raises lactate

When pyruvate cannot be efficiently oxidized, more is reduced to lactate. ATP supply to highly oxidative tissues falls. Arsenic can interfere with lipoate-dependent enzyme complexes. These are mechanism comparisons, not diagnostic conclusions from lactate alone.

PDH component	Role	Cofactor link
E1, pyruvate dehydrogenase	Decarboxylates pyruvate and transfers the hydroxyethyl equivalent	TPP
E2, dihydrolipoyl transacetylase	Oxidizes/transfers the acetyl group to CoA using a swinging lipoyl arm	Lipoate and CoA
E3, dihydrolipoyl dehydrogenase	Reoxidizes reduced lipoate and transfers reducing equivalents onward	FAD then NAD ⁺

Arsenite inhibits lipoate-dependent complexes, whereas arsenate can substitute for phosphate in the GAP-dehydrogenase reaction and bypass capture of energy at phosphoglycerate kinase. Distinguish these chemical forms and sites rather than saying every arsenic question has the same mechanism.

Sources for this section

Biochemistry Metabolism (Part 1) Notes - Hawraa.pdf - PDF pp. 1-7; all handwritten pages visually inspected

Biochemistry metabolism notes (pt 2) slides 27-80 .pdf - PDF pp. 1-6

IMG_4406.transcript.txt - entire transcript

IMG_4407.transcript.txt - entire transcript

Biochemistry Metabolism Recorded Lecture - 1h04m20s.transcript.txt - entire transcript

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch. 9 pp. 209-223

Respiratory chain and transparent ATP accounting

SOURCE BASIS: lecture + book

Electrons from NADH enter complex I. Electrons from succinate oxidation enter through complex II. Coenzyme Q transfers electrons to III, cytochrome c to IV, and IV reduces oxygen to water. Complexes I, III and IV pump protons; II does not. ATP synthase uses proton return to the matrix. The chain and phosphorylation normally run together because the proton gradient restrains further pumping when it is not dissipated.

Perturbation	Electron flow / oxygen use	ATP and gradient
Respiratory-chain inhibitor	Electron flow and oxygen consumption fall downstream of the block	ATP production falls.
ATP synthase inhibitor	Backpressure from the gradient slows respiratory electron flow	ATP falls while the proton gradient initially rises.
Uncoupler or UCP1	Protons return without ATP synthase; respiration can increase when substrate is available	Less ATP per oxygen consumed; energy appears as heat.

Source per glucose	Products	ATP equivalents at NADH 2.5, FADH2 1.5
Glycolysis	2 net ATP + 2 cytosolic NADH	2 + 3 or 5, depending on shuttle
Two PDH reactions	2 NADH	5
Two TCA turns	6 NADH + 2 FADH2 + 2 GTP	15 + 3 + 2 = 20
Complete oxidation	Sum of all stages	30 or 32 ATP, theoretical bookkeeping

Edition convention matters

The inspected 6th edition uses older values of three ATP per NADH and two per FADH2 in some chapters. The lecture uses 30-32 ATP per glucose. This review uses 2.5/1.5 for modern worked calculations and states when an old-book answer differs. Actual cellular yields vary with transport costs, proton leak and physiological conditions.

Agent/site	Direct effect	Predict the downstream response
Rotenone or amytal: complex I	Blocks NADH-derived electron transfer toward CoQ	Succinate-derived entry through complex II can bypass the I block in an appropriate experimental system.
Antimycin A: complex III	Blocks electron transfer through cytochrome bc1	NADH- and succinate-derived paths converge before this block, so simply changing between them does not bypass III.
Cyanide, carbon monoxide, hydrogen sulfide or azide: complex IV	Interferes with terminal reduction of oxygen	Upstream carriers tend to become reduced; oxidative ATP supply falls.

Agent/site	Direct effect	Predict the downstream response
Oligomycin: ATP-synthase Fo channel	Blocks proton return through synthase	ATP synthesis falls; the accumulating gradient slows respiration.
2,4-dinitrophenol or UCP1	Dissipates gradient without ATP capture	Electron transfer can accelerate but ATP efficiency falls and heat rises.

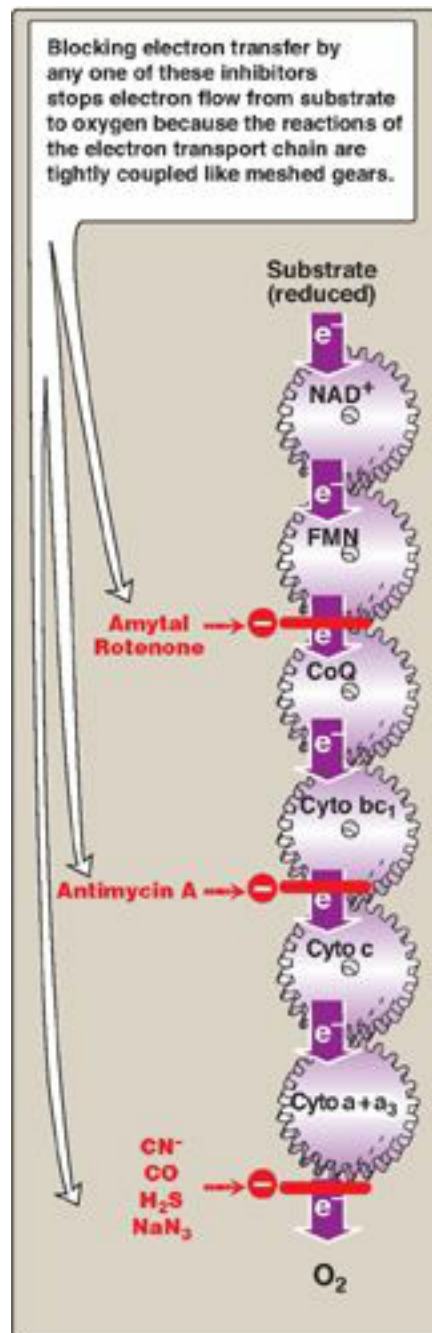


Figure 6.10 from the local book: site-specific inhibitor map. Use it to locate the block, then predict oxidation states on either side; do not merely memorize a poison name.

Identify: Complex I; Coenzyme Q; Complex III; Cytochrome c; Complex IV

Figure source: Lippincott Biochemistry 6e, PDF p. 149, extracted source diagram

Component	Recognize	Reasoning
Complex I	NADH dehydrogenase: FMN and Fe-S centers	Accepts NADH electrons; pumps H ⁺ .
Complex II	Succinate-Q reductase: FAD and Fe-S centers	No proton pumping; feeds electrons to Q.
Complex III / IV	Cytochromes b/c1 / cytochromes a and a ₃ with copper centers	Cytochrome c transfers one electron; Q can carry two electrons and protons. O ₂ is the final acceptor at IV.
ATP synthase / ANT	Oligomycin blocks F ₀ ; atractyloside blocks the ADP/ATP translocator	ANT exchanges cytosolic ADP for matrix ATP; it is not itself an ETC complex.
Proton motive force	Membrane voltage plus a pH gradient	The inner membrane must restrict nonspecific ion flow. A freely permeable membrane cannot sustain coupled ATP synthesis.

Electron flow generally proceeds toward more positive reduction potential. Of FMN and the downstream carriers used in these papers, FMN is the relatively weak oxidant; NADH upstream has still lower reduction potential. Do not interpret the MCQ as claiming FMN is the lowest-potential substance in all biology. Dehydrogenases transfer electrons to cofactors and do not all use molecular oxygen directly.

Accounting question	Modern convention	Historical convention
One acetyl-CoA in TCA	$3(2.5) + 1(1.5) + 1 = 10$ ATP	$3(3) + 1(2) + 1 = 12$ ATP
Alpha-ketoglutarate to succinate	One NADH + one GTP = 3.5 ATP equivalents	4 ATP equivalents
Highest phosphate-transfer potential among usual options	PEP exceeds 1,3-BPG, phosphocreatine and ATP	Use the actual listed compounds; ATP is not the highest-energy phosphate.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch. 6 pp. 135-158; Ch. 9 p.220

Biochemistry Metabolism Recorded Lecture - 1h04m20s.transcript.txt - final TCA explanation and 30-32 ATP discussion

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition - PDF pp. 143-155

Fructose, galactose, mannose and the polyol pathway

SOURCE BASIS: lecture + book

Hepatic fructokinase consumes ATP to form fructose-1-phosphate. Aldolase B cleaves this to DHAP and glyceraldehyde, and triose kinase phosphorylates glyceraldehyde to GAP. Entry below PFK-1 bypasses that major glycolytic control point. Hexokinase can form fructose-6-phosphate in some tissues but generally prefers glucose. Fructokinase deficiency causes relatively benign essential fructosuria. Aldolase B deficiency traps phosphate as fructose-1-phosphate, reducing ATP and impairing glycogenolysis/gluconeogenesis: hereditary fructose intolerance becomes apparent after fructose, sucrose or sorbitol exposure.

Galactose is the C4 epimer of glucose. Galactokinase makes galactose-1-phosphate. GALT transfers a uridyl group from UDP-glucose: galactose-1-phosphate + UDP-glucose produces UDP-galactose + glucose-1-phosphate. UDP-galactose 4-epimerase regenerates UDP-glucose. Phosphoglucomutase converts glucose-1-phosphate to glucose-6-phosphate. The reconstructed notes abbreviate this pathway so heavily that the UDP-glucose partner and released glucose-1-phosphate can be missed.

Defect/pathway	Biochemical consequence	Distinguishing pattern
Galactokinase deficiency	Galactose diverted to galactitol	Cataracts with a milder systemic picture.
Classic GALT deficiency	Galactose-1-phosphate accumulation and galactitol production	Illness after milk feeding, liver dysfunction and cataracts; different from lactose intolerance.
Lactase deficiency	Lactose remains in intestinal lumen	Osmotic diarrhea and bacterial fermentation, without a primary intracellular galactose-enzyme defect.
Aldose reductase	Glucose reduced to sorbitol using NADPH	Sorbitol accumulates in tissues with limited sorbitol dehydrogenase during hyperglycemia.
Sorbitol dehydrogenase	Sorbitol to fructose, making NADH	Prominent in tissues such as liver and seminal vesicles.
Mannose	Hexokinase makes mannose-6-P, then isomerization to fructose-6-P	Connects glycoprotein carbohydrate metabolism to glycolysis.

Check yourself: Why is the intact GALT reaction more informative than writing galactose-1-P to UDP-galactose?

The intact equation shows both UDP-glucose consumption and glucose-1-phosphate release. It explains the transferase name, how glucose carbon enters mainstream metabolism, and why epimerase is needed to replenish the UDP-glucose pool.

Lactose synthesis occurs in the Golgi of the lactating mammary gland. Lactose synthase combines a broadly expressed beta-galactosyltransferase with alpha-lactalbumin, whose expression is stimulated by prolactin. Alpha-lactalbumin changes the transferase substrate preference so it transfers galactose from UDP-galactose to glucose, producing lactose plus UDP. Without this modifying protein, the transferase preferentially forms N-acetyllactosamine using N-acetylglucosamine. This is a synthesis mechanism, separate from intestinal lactase digestion.

Differentiate GLUT5-related fructose malabsorption (impaired intestinal entry) from hereditary fructose intolerance (aldolase B deficiency inside cells). In the latter, fructose-1-phosphate traps phosphate, ATP falls, AMP degradation can raise urate, and hepatic glucose output is impaired. Classic galactosemia similarly accumulates galactose-1-phosphate and may cause hypoglycemia and hepatic injury; galactokinase deficiency is usually dominated by cataracts. Aldose reductase consumes NADPH to make sorbitol from glucose or galactitol from galactose. The lens has inadequate sorbitol dehydrogenase to clear the load. Polyol accumulation and nonenzymatic advanced glycation end products are different processes, not interchangeable carbohydrate names.

Sources for this section

Biochemistry metabolism notes (pt 2) slides 27-80 .pdf - PDF pp. 1-2

Biochemistry Metabolism Recorded Lecture - 1h04m20s.transcript.txt - opening fructose/galactose section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch. 12 pp. 262-276

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition - PDF pp. 230-245

Glycogen structure, synthesis, degradation and control

SOURCE BASIS: lecture + book

Glycogen is a branched glucose polymer with alpha-1,4 chains and alpha-1,6 branch points. Its numerous nonreducing ends permit rapid simultaneous synthesis and breakdown. Glycogenin creates the primer. UDP-glucose supplies activated glucose to glycogen synthase, which makes alpha-1,4 links. Branching enzyme transfers a chain segment to form an alpha-1,6 branch. Adding a free glucose to glycogen consumes an ATP for glucose phosphorylation and a UTP equivalent for UDP-glucose formation.

Glycogen phosphorylase uses inorganic phosphate, not water, to remove alpha-1,4-linked residues as glucose-1-phosphate. It requires PLP and stops four residues before a branch. The debranching transferase moves three residues, then alpha-1,6-glucosidase releases the remaining branch-point glucose as free glucose. Most residues therefore emerge already phosphorylated; the small branch-point fraction is free glucose. In muscle, glycolysis of a glycogen-derived glucose-6-phosphate can net three ATP to lactate because the hexokinase ATP cost was avoided.

Situation	Liver	Skeletal muscle
Purpose of glycogen	Maintain blood glucose	Provide fuel for contraction
Glucose-6-phosphatase	Present: can release free glucose	Absent: G6P retained for local metabolism
Glucagon response	Activates cAMP/PKA glycogenolysis	No direct physiological glucagon response
Epinephrine	Promotes glycogen mobilization	Promotes glycogen mobilization
Calcium / AMP	Calcium can activate phosphorylase kinase; glucose signals abundance	Contraction-derived Ca ²⁺ activates phosphorylase kinase; AMP activates phosphorylase

Reciprocal phosphorylation

Phosphorylation activates phosphorylase kinase and glycogen phosphorylase while inhibiting glycogen synthase. Insulin-promoted dephosphorylation reverses the storage/mobilization balance. Always name the enzyme before deciding whether phosphorylation activates it.

Storage disease	Defect	Mechanistic clinical contrast
Von Gierke, type I	Glucose-6-phosphatase system	Severe fasting hypoglycemia because final glucose release fails in both glycogenolysis and gluconeogenesis.
Pompe, type II	Lysosomal acid alpha-glucosidase	Lysosomal glycogen accumulation, cardiomyopathy/myopathy; different compartment from cytosolic degradation.
Cori, type III	Debranching enzyme	Limit-dextrin accumulation and incomplete glycogen degradation.
Andersen, type IV	Branching enzyme	Abnormally poorly branched glycogen, especially damaging to liver.
McArdle, type V	Muscle glycogen phosphorylase	Exercise intolerance, cramps and possible myoglobinuria; liver glucose release preserved.
Hers, type VI	Liver glycogen phosphorylase	Hepatic glycogenolysis impaired; usually milder fasting phenotype than type I.

Calcium activates phosphorylase kinase by binding its calmodulin subunit, linking contraction to fuel release even before a large change in ATP. Muscle glycogen phosphorylase is allosterically activated by AMP and inhibited by ATP and glucose-6-phosphate. Hepatic phosphorylase responds to glucose availability. In synthesis, UDP-glucose pyrophosphorylase activates glucose-1-phosphate using UTP; pyrophosphate hydrolysis favors the reaction. Glycogenin autoglucosylates a tyrosine hydroxyl to start the primer.

Von Gierke disease combines fasting hypoglycemia with hepatomegaly, lactic acidosis, hyperlipidemia and often hyperuricemia, not hypouricemia. McArdle disease is a muscle defect: exercise cramps and myoglobinuria can occur, but hepatomegaly is not its defining pattern. In plant carbohydrate synthesis, ADP-glucose supplies starch, whereas UDP-glucose supplies cellulose; neither should be silently substituted for the other when a paper omits the correct option.

Sources for this section

Biochemistry metabolism notes (pt 2) slides 27-80 .pdf - PDF p.3

Biochemistry Metabolism Recorded Lecture - 1h04m20s.transcript.txt - glycogen and GLUT2/GLUT4 discussion

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.11 pp.240-261

carbohydrate metabolism MD Kish.pdf - PDF pp. 35-37 and carbohydrate biosynthesis slides

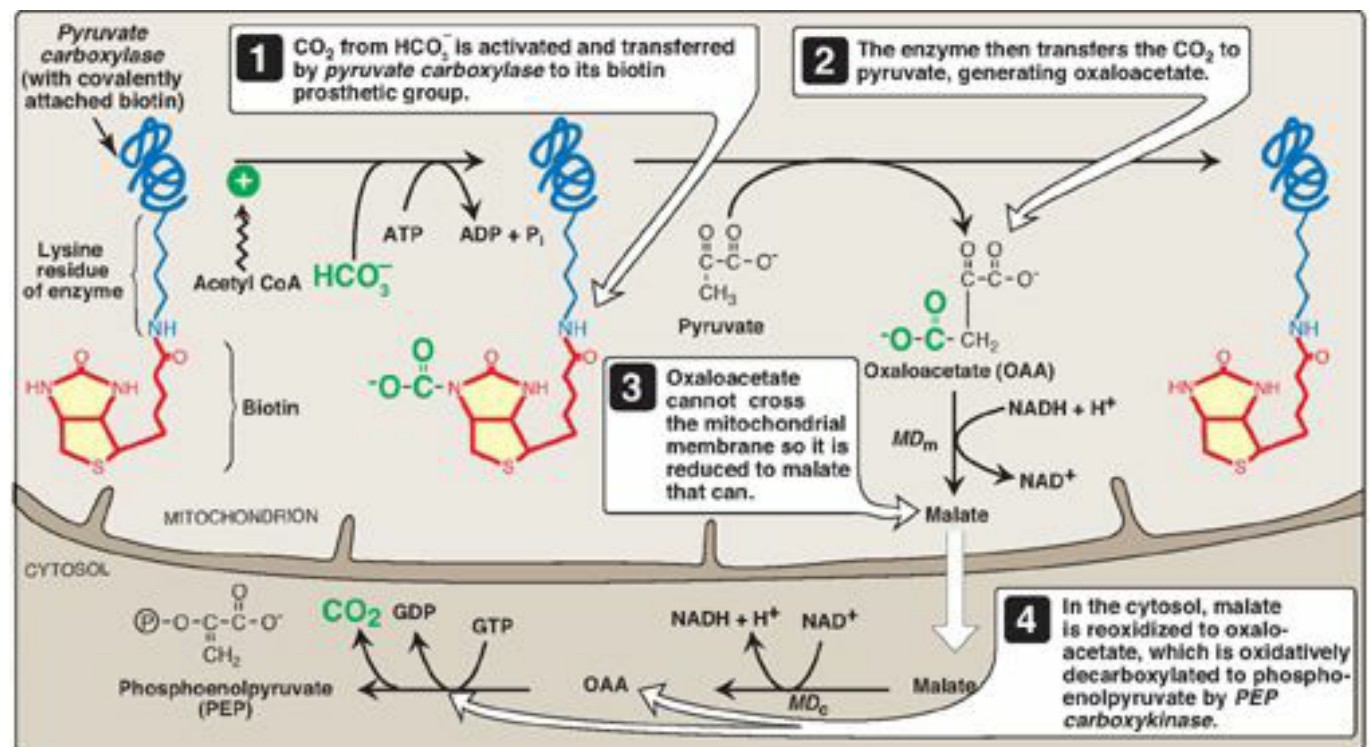
Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition - PDF pp. 246-265

Gluconeogenesis, redox transfer and substrate limits

SOURCE BASIS: lecture + book

Liver is the main fasting glucose producer. Kidney cortex becomes increasingly important in prolonged fasting. Gluconeogenesis uses seven reversible glycolytic reactions plus bypasses for the three irreversible steps. Major substrates are lactate, glycerol and glucogenic amino-acid carbon skeletons. The glycerol backbone of TAG can yield glucose, but acetyl-CoA from its even-chain fatty acids cannot produce net glucose. Oxaloacetate is regenerated rather than created net when acetyl-CoA merely turns the TCA cycle.

Bypass	Enzyme/location	Requirement
Pyruvate to oxaloacetate	Pyruvate carboxylase, mitochondrial matrix	ATP, biotin and bicarbonate; acetyl-CoA activates.
Oxaloacetate to PEP	PEP carboxykinase; cytosolic and mitochondrial isoforms	GTP; releases the previously added CO ₂ .
Fructose-1,6-bisP to fructose-6-P	Fructose-1,6-bisphosphatase	Hydrolysis, not ATP generation. Inhibited by AMP and fructose-2,6-bisP.
Glucose-6-P to glucose	Glucose-6-phosphatase system in ER	Permits liver/kidney glucose export; absent from muscle.



Source diagram of the pyruvate-carboxylase and cytosolic-PEPCK route: track carbon, biotin and the malate shuttle separately. Other substrate-dependent routes may use mitochondrial PEPCK.

Identify: Pyruvate carboxylase; Biotin; Oxaloacetate; Malate; Cytosolic NADH; PEPCK

Figure source: Lippincott Biochemistry 6e, PDF p. 229, extracted source diagram

Oxaloacetate cannot directly cross the inner mitochondrial membrane. In the route emphasized by the lecture, reduction to malate allows export; cytosolic reoxidation restores oxaloacetate and provides NADH needed later in glucose synthesis. Lactate oxidation already supplies cytosolic NADH,

so route selection can differ according to precursor. Glycerol enters via glycerol-3-phosphate then DHAP, downstream of the pyruvate bypass. From two pyruvates to one glucose, the requirement is four ATP, two GTP and two NADH.

Alcohol, fasting and hypoglycemia

Ethanol oxidation raises hepatic NADH/NAD⁺. This favors pyruvate-to-lactate and oxaloacetate-to-malate conversion, reducing available gluconeogenic intermediates. The risk of hypoglycemia is especially important when glycogen stores are depleted. Glycerol is not restricted to glycolysis during starvation: the hormonal setting favors its contribution to glucose synthesis.

Corrections to the newly recovered carbohydrate slides

The carbohydrate deck p. 53 loosely places gluconeogenesis in liver and muscle. Muscle can metabolize lactate but lacks the glucose-6-phosphatase system needed for net free-glucose export. Liver and kidney are the relevant glucose-producing organs. Page 54 also overgeneralizes conversion of all precursors to pyruvate: glycerol enters at DHAP and several amino acids enter through other intermediates.

Sources for this section

Biochemistry metabolism notes (pt 2) slides 27-80 .pdf - PDF pp.4-5

Biochemistry Metabolism Recorded Lecture - 1h04m20s.transcript.txt - gluconeogenesis, malate and energy discussion

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.10 pp.224-239

carbohydrate metabolism MD Kish.pdf - PDF pp. 38-54; source wording compared with Ferrier ch. 10

Pentose phosphate pathway and antioxidant defense

SOURCE BASIS: lecture + book

The cytosolic pentose phosphate pathway serves two demands: NADPH and ribose-5-phosphate. Its oxidative phase converts glucose-6-phosphate to ribulose-5-phosphate, releasing CO₂ and producing two NADPH. G6PD controls entry and responds to the NADP⁺/NADPH balance. The nonoxidative phase reversibly exchanges sugar carbons through transketolase and transaldolase, linking pentoses to fructose-6-phosphate and GAP. Transketolase requires TPP. No ATP is directly made by the pathway.

Cellular need	Pathway strategy	Reasoning
Ribose exceeds NADPH demand	Reverse nonoxidative reactions from glycolytic intermediates	Build ribose without obligatory oxidative NADPH production.
NADPH and ribose both needed	Oxidative phase followed by pentose isomerization	A biosynthetic cell can use both products.
NADPH exceeds ribose demand	Recycle pentoses into glycolytic intermediates and regenerate hexose phosphate	Keep extracting reducing power without accumulating unused ribose.

Glutathione peroxidase uses reduced glutathione to remove peroxides. Glutathione reductase uses NADPH to replenish reduced glutathione. RBCs have no mitochondria and depend strongly on the PPP for this defense. G6PD deficiency predisposes to oxidative hemolysis, with damaged hemoglobin forming Heinz bodies and splenic removal producing bite cells. Oxidant exposure, fava beans or infection can trigger episodes.

The same reductant can protect cells or help kill microbes: phagocyte NADPH oxidase forms superoxide, superoxide dismutase forms hydrogen peroxide, and myeloperoxidase uses peroxide and chloride to form hypochlorous acid. NADPH oxidase failure causes chronic granulomatous disease. Nitric-oxide synthase uses arginine, oxygen and NADPH to form NO, linking the pathway to vascular signaling.

Sources for this section

Biochemistry Metabolism (Part 1) Notes - Hawraa.pdf - PDF p.1: phosphogluconate pathway mentioned

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.13 pp.277-297

Dietary lipid digestion: from meal to absorbable products

SOURCE BASIS: lecture + book

Most dietary lipid is TAG. Phospholipids, free cholesterol, cholesteryl esters and fat-soluble vitamins accompany it. Hydrophobic lipid must be presented at a lipid-water interface before water-soluble digestive enzymes can work efficiently. Bile salts emulsify lipid droplets and stabilize smaller particles; they do not hydrolyze ester bonds. Peristalsis supplies mechanical mixing.

Agent	Site/activation	Substrate and result
Lingual and gastric lipases	Work in acidic stomach conditions; important for milk fat and when pancreatic function is poor	Partial TAG hydrolysis, particularly involving shorter fatty-acid chains.
Pancreatic lipase + colipase	Small intestine. Trypsin activates procolipase. Colipase anchors lipase at the interface despite bile salts	Removes TAG fatty acids at positions 1 and 3, leaving 2-monoacylglycerol + two fatty acids.
Cholesterol esterase	Pancreatic secretion; bile salts enhance activity	Cholesteryl ester to free cholesterol + fatty acid.
Phospholipase A2	Pancreatic proenzyme activated by trypsin	Removes fatty acid at phospholipid position 2, producing lysophospholipid; further hydrolysis can follow.
CCK	Intestinal response to lipid and partially digested protein	Gallbladder contraction and pancreatic enzyme secretion; slows gastric emptying.
Secretin	Intestinal response to acidic chyme	Pancreatic bicarbonate secretion creates a suitable pH for pancreatic enzymes.

Do not turn a source generalization into a chemical rule

TAG is glycerol esterified to three fatty acids, not cholesterol plus three fatty acids. Oils and fats differ largely in melting behavior, influenced by saturation and chain length. Plant versus animal origin is not an absolute solid/liquid rule.

Source-specific intake values

The Esmaeili 1.1 transcript, lines 99-100, describes about 78 g dietary lipid per day. Lippincott 6e PDF p.328 describes about 81 g/day for U.S. adults in that edition. Both emphasize more than 90% as TAG. These are attributed population descriptions, not a universal daily requirement, and should not be averaged into an invented value.

Relatively acid-stable lingual and gastric lipases are particularly helpful for short/medium-chain TAG in milk. Their contribution matters in neonates and can partly support lipid digestion when pancreatic function is impaired. This does not mean they fully replace pancreatic digestion of all dietary long-chain fat.

Orlistat inhibits gastric and pancreatic lipases. Gallbladder bile supplies bile salts but is not the source of pancreatic enzymes; conversely, pancreatic juice does not supply bile salts. Aldolase is a glycolytic enzyme, not a pancreatic digestive hydrolase. CCK promotes enzyme release and gallbladder contraction while slowing gastric emptying; secretin mainly promotes pancreatic bicarbonate release.

Sources for this section

Lipid metabolism Esmaeili 1.1.transcript.txt - entire transcript

Lipid metabolism Esmaeili 1.2.transcript.txt - entire transcript

Lipid metabolism1 Esmaeili 1.3.transcript.txt - entire transcript

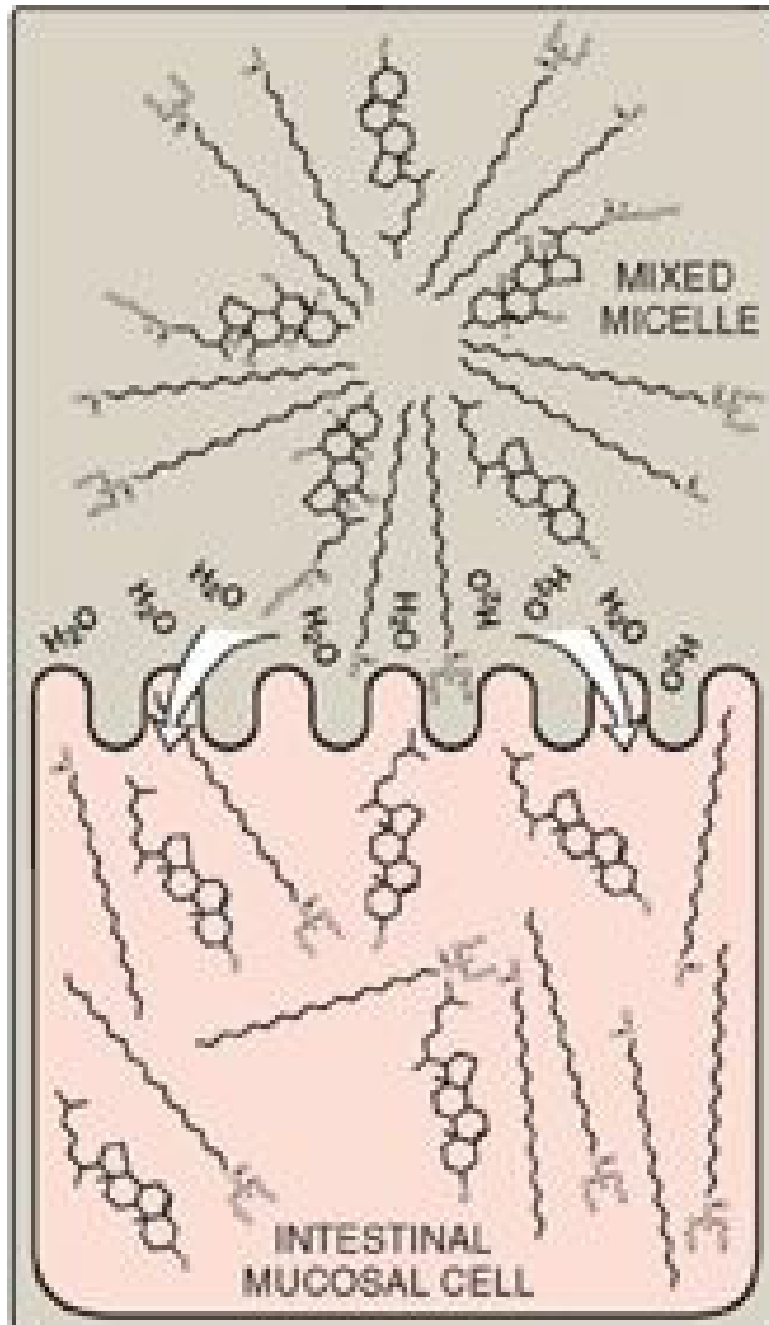
Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.15 pp.326-331

Lipid metabolism 1.pdf - PDF pp. 4-19

Micelles, enterocytes, chylomicrons and remnant uptake

SOURCE BASIS: lecture + book

Mixed micelles contain bile salts, fatty acids, 2-monoacylglycerol, cholesterol and fat-soluble vitamins. They facilitate movement through the unstirred aqueous layer to the enterocyte brush border. The lipid contents enter the mucosal cell; the entire micelle is not simply converted into a chylomicron. Most bile salts remain in the lumen until reabsorption later, mainly in terminal ileum.



The source image shows the mixed micelle at the intestinal brush border. Distinguish its luminal transport role from intracellular assembly of a protein-containing chylomicron.

Identify: Mixed micelle; Bile-salt surface; Lipid contents; Brush border; Intestinal mucosal cell

Figure source: Lippincott Biochemistry 6e, PDF p. 333, extracted source diagram

Long-chain fatty acids are activated to acyl-CoA at an ATP-to-AMP cost. Acyltransferases rebuild TAG from 2-monoacylglycerol through DAG. Cholesterol is esterified intracellularly by ACAT. TAG and cholesteryl esters occupy the hydrophobic particle core, while phospholipid, free cholesterol and apoprotein form the surface. ApoB-48 and microsomal triglyceride transfer protein are required for chylomicron assembly. Chylomicrons leave by exocytosis into lacteals, then lymph and the thoracic duct, reaching venous blood at the left venous angle. Short- and medium-chain fatty acids largely reach portal blood without the micelle/chylomicron route.

In plasma, chylomicrons acquire apoC-II and apoE from HDL. ApoC-II activates lipoprotein lipase on capillary endothelium of adipose and muscle. Released fatty acids enter tissues for storage or oxidation. Glycerol travels especially to liver, whose glycerol kinase can make glycerol-3-phosphate. Adipose has little glycerol-kinase activity and relies heavily on glucose-derived DHAP for the glycerol

backbone of storage TAG. After TAG depletion, the cholesterol-rich remnant is taken up by liver through apoE-dependent recognition.

Failure	Immediate consequence	Clinical pattern
Pancreatic insufficiency or blocked ducts	Digestive enzymes fail to reach lipids	Steatorrhea and fat-soluble vitamin malabsorption.
Cholestasis or impaired bile-salt availability	Emulsification/micellar transport inadequate	Fat malabsorption; distinguish from a plasma lipoprotein clearance defect.
LPL or apoC-II deficiency	Chylomicron TAG cannot be efficiently cleared	Fasting chylomicronemia and severe hypertriglyceridemia; pancreatitis risk.
ApoE-related remnant-clearance defect	Remnants accumulate	Type III dysbetalipoproteinemia, involving both chylomicron and VLDL remnants.
ApoB-particle assembly failure	Intestinal lipid export impaired	Fat malabsorption and deficient delivery of fat-soluble vitamins.

CFTR explanation requires tissue specificity

Cystic fibrosis can obstruct pancreatic ducts with thick secretions and cause exocrine insufficiency. CFTR physiology differs between secretory epithelia and sweat ducts: sweat duct failure impairs salt reabsorption, so do not repeat the transcript claim that every affected tissue simply secretes less chloride in the same way.

Sources for this section

Lipid metabolism Esmaeili 1.1.transcript.txt - entire transcript

Lipid metabolism Esmaeili 1.2.transcript.txt - entire transcript

Lipid metabolism1 Esmaeili 1.3.transcript.txt - entire transcript

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.15 pp.331-337; Ch.18 pp.420-425

Fatty-acid synthesis and TAG storage

SOURCE BASIS: book extension

Fatty-acid synthesis predominates when carbon and energy are abundant. Mitochondrial acetyl-CoA cannot directly cross the inner membrane, so citrate carries its acetyl units to the cytosol. ATP-citrate lyase releases cytosolic acetyl-CoA. Acetyl-CoA carboxylase uses ATP, bicarbonate and biotin to make malonyl-CoA. This is the regulated committed step. Citrate activates ACC, long-chain acyl-CoA inhibits it, and AMPK-mediated phosphorylation inhibits it during energy shortage. Insulin favors activation and increases lipogenic enzyme expression.

Fatty-acid synthase holds the growing chain on its ACP domain using phosphopantetheine derived from vitamin B5. One acetyl primer and seven malonyl units yield palmitate after seven cycles of condensation, reduction, dehydration and reduction. Each cycle consumes two NADPH. Seven ATP form the malonyl-CoA units, and fourteen NADPH supply reducing power, excluding upstream citrate-shuttle costs. The added bicarbonate carbon is released during condensation and is not retained in palmitate. The thioesterase releases free palmitate, not palmitoyl-CoA.

- PPP and cytosolic malic enzyme supply NADPH. Malic enzyme converts malate to pyruvate with oxidative decarboxylation, unlike NADH-linked malate dehydrogenase.
- ER enzymes elongate and desaturate fatty acids. Humans cannot create the omega-6 and omega-3 parent structures de novo: linoleic and alpha-linolenic acids are essential.
- TAG synthesis joins activated fatty acids to glycerol-3-phosphate through phosphatidic acid and DAG. Liver can use glycerol directly and export TAG in VLDL. Adipose predominantly obtains glycerol-3-phosphate from glycolysis and stores TAG.
- Insulin coordinates GLUT4-mediated glucose uptake, glycerol-backbone supply and adipose lipid storage. During fasting, reduced insulin and catecholamine signals favor lipolysis.

Malonyl-CoA separates incompatible pathways

Malonyl-CoA is a synthesis substrate and inhibits CPT-I. Therefore newly synthesized fatty acids are not simultaneously imported into mitochondria for rapid oxidation. The carnitine shuttle, not beta-oxidation chemistry itself, is the controlled entry gate.

Humans can introduce a delta-9 double bond but cannot synthesize the omega-6/omega-3 parent fatty acids through delta-12/delta-15 desaturation. Metformin can favor AMPK-mediated ACC inhibition; cAMP-linked counterregulatory signaling also favors inhibition. These are regulatory pathways, not proof of direct binding of those signals to the ACC catalytic site. Malonyl-CoA is needed for fatty-acid synthesis, not the cholesterol pathway.

Sources for this section

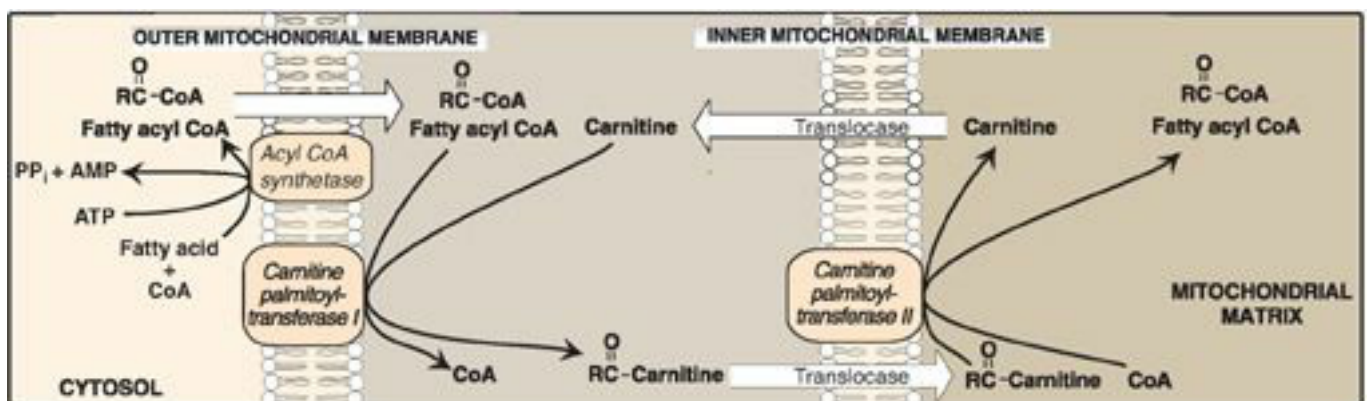
Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.16 pp.340-355

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition - PDF pp. 340-351

Lipolysis, carnitine transport and beta-oxidation

SOURCE BASIS: book extension

Adipose TAG mobilization involves sequential lipases, including adipose triglyceride lipase and hormone-sensitive lipase. Released nonesterified fatty acids travel bound to albumin. Activation by acyl-CoA synthetase consumes ATP to AMP plus pyrophosphate, costing two high-energy phosphate equivalents. Long-chain acyl groups cross the inner mitochondrial membrane as acylcarnitine: CPT-I forms acylcarnitine, a translocase exchanges it across the membrane, and CPT-II regenerates acyl-CoA in the matrix.



Carnitine shuttle source diagram. Identify the outer-membrane CPT-I, inner-membrane translocase and matrix-side CPT-II. Acyl groups cross; the original cytosolic CoA does not.

Identify: CPT-I; Acylcarnitine; Translocase; CPT-II; Matrix acyl-CoA

Figure source: Lippincott Biochemistry 6e, PDF p. 357, extracted source diagram

Repeated step	Enzyme/function	Product
First oxidation	Acyl-CoA dehydrogenase	trans-2-enoyl-CoA and FAD-linked reducing equivalents
Hydration	Enoyl-CoA hydratase	L-3-hydroxyacyl-CoA
Second oxidation	3-hydroxyacyl-CoA dehydrogenase	3-ketoacyl-CoA and NADH
Thiolysis	Beta-ketothiolase plus CoA	Acetyl-CoA and acyl-CoA shortened by two carbons

For an even saturated n-carbon fatty acid, $n/2$ acetyl-CoA units require $n/2$ minus one beta-oxidation cycles. Palmitate (16:0) gives eight acetyl-CoA, seven NADH and seven FADH₂. At 2.5 ATP/NADH and 1.5 ATP/FADH₂: $8 \times 10 + 7 \times 2.5 + 7 \times 1.5 = 108$ ATP before activation, or 106 net from free palmitate. Starting with palmitoyl-CoA omits that activation charge. The old 6e calculation gives 131 gross and 129 net using 3/2 ratios. State the convention rather than treating these as different carbon pathways.

Special case	Additional requirement	Exam distinction
Odd-chain fatty acid	Propionyl-CoA carboxylase (biotin), epimerase, methylmalonyl-CoA mutase (B12)	Succinyl-CoA can contribute net glucogenic carbon.
Unsaturated fatty acid	Enoyl-CoA isomerase; polyunsaturated substrates may also need NADPH-dependent reductase	Yield is lower than the corresponding saturated chain; specify double-bond arrangement before calculating.
Very-long-chain fatty acid	Initial peroxisomal shortening	Initial oxidation transfers electrons to oxygen, creating H ₂ O ₂ rather than respiratory-chain ATP.
Phytanic acid	Peroxisomal alpha-oxidation	A beta-methyl branch obstructs ordinary beta-oxidation; Refsum disease is the classic defect.
Omega oxidation	ER pathway produces dicarboxylic acids	Can increase when mitochondrial beta-oxidation is impaired.

Hypoketotic hypoglycemia

MCAD deficiency or a hepatic long-chain fatty-acid entry defect impairs fuel oxidation and reduces ketone production during fasting. Glucose demand remains high while hepatic energy supply for gluconeogenesis falls. This differs from the marked ketosis expected in severe insulin deficiency.

For saturated even chains, C14 requires six cycles and C18 eight. Water is consumed during hydration, not generated as a net beta-oxidation product; neither NADPH nor malonyl-CoA is a product. In conventional odd-chain bookkeeping, C17 gives seven acetyl-CoA plus propionyl-CoA. Counting propionyl-CoA to oxaloacetate gives 5 ATP equivalents minus 1 ATP for carboxylation; total C17 yield is $7(10)+7(4)+4-2=100$, compared with 106 for C16. C13 similarly gives 72 versus 78 for C12. This is a stated bookkeeping endpoint, not complete oxidation of all propionyl carbon to CO₂ or a universal calorimetry rule. Unsaturated fatty acids yield less than a saturated chain of the same length; multiple double bonds further alter the accounting.

Zellweger-spectrum peroxisome-biogenesis disorders impair very-long-chain fatty-acid oxidation and other peroxisomal functions, not primarily short-chain mitochondrial oxidation. Refsum disease impairs alpha-oxidation of phytanic acid. Carnitine is necessary for long-chain mitochondrial import, but knowing that mechanism does not establish benefits from taking carnitine supplements in an otherwise replete person.

Sources for this section

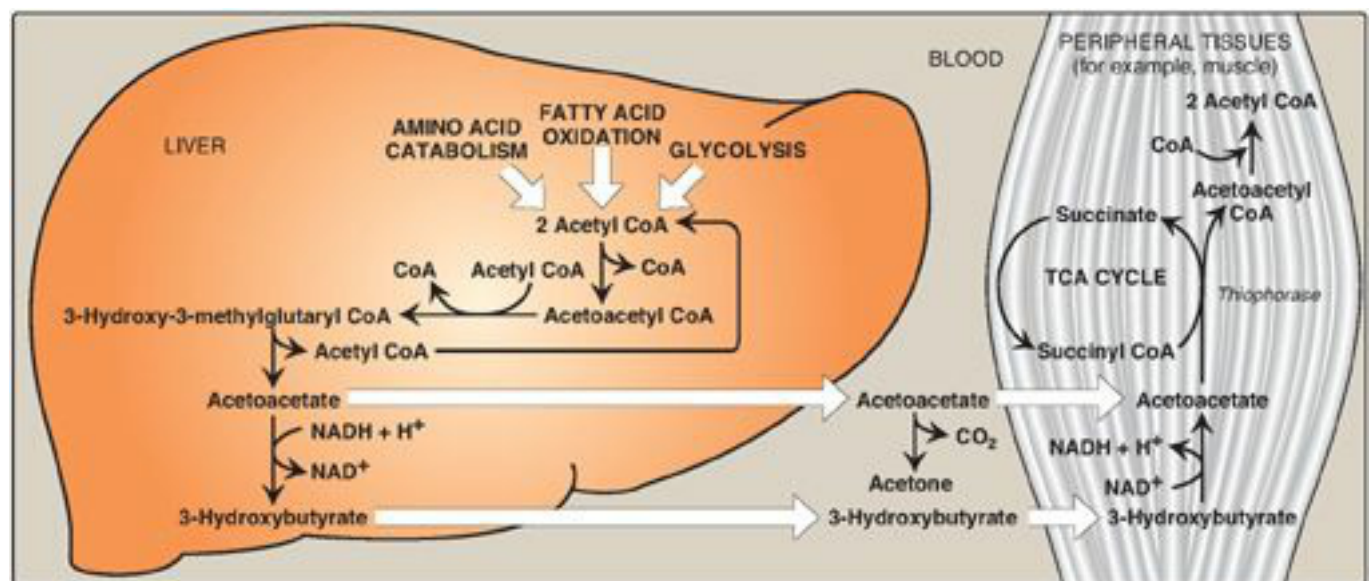
Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.16 pp.355-363; old ATP accounting p.359

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition - PDF pp. 352-366

Ketogenesis and ketone utilization

SOURCE BASIS: book extension

During fasting, fatty acids reach liver and generate abundant acetyl-CoA and NADH. Oxaloacetate is diverted toward gluconeogenesis and its reduction to malate is favored by the redox state. Acetyl-CoA therefore exceeds the capacity for complete TCA oxidation. Liver mitochondria condense acetyl-CoA to acetoacetyl-CoA, add another acetyl-CoA through mitochondrial HMG-CoA synthase, then use HMG-CoA lyase to form acetoacetate. Acetoacetate can be reduced to beta-hydroxybutyrate or spontaneously decarboxylate to acetone.



Ketone synthesis in liver and utilization in peripheral mitochondria. Note the separate HMG-CoA pathway and the peripheral thiophorase reaction.

Identify: Mitochondrial HMG-CoA; Acetoacetate; Beta-hydroxybutyrate; Thiophorase/SCOT; Acetoacetyl-CoA

Figure source: Lippincott Biochemistry 6e, PDF p. 366, extracted source diagram

Ketones are water-soluble fuels. Extrahepatic mitochondria oxidize beta-hydroxybutyrate to acetoacetate, then SCOT (thiophorase) transfers CoA from succinyl-CoA. Thiolase generates two acetyl-CoA for oxidation. Liver lacks SCOT and therefore exports rather than consumes ketones. Mature RBCs cannot use them because they lack mitochondria. Brain ketone use increases during prolonged fasting and reduces, but does not abolish, the need for glucose and amino-acid-derived gluconeogenesis.

Three chemically different ketone bodies

Beta-hydroxybutyrate is chemically a hydroxy acid, despite the physiological name ketone body. Acetone does not contribute meaningful ATP. A high hepatic NADH/NAD⁺ ratio favors beta-hydroxybutyrate relative to acetoacetate. Unrestrained ketoacid production can consume bicarbonate and cause high-anion-gap acidosis.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.16 pp.364-369; Ch.24 pp.600-608

Membrane lipids, sphingolipidoses and eicosanoids

SOURCE BASIS: book extension

Class	Structural logic	Function
Glycerophospholipid	Glycerol with fatty acids and phosphate-linked head group	Membrane structure, signaling and surfactant.
Sphingomyelin	Ceramide plus phosphocholine	Myelin and membrane phospholipid, not glycerol based.
Cerebroside	Ceramide plus one sugar	Neutral glycosphingolipid.
Ganglioside	Ceramide plus oligosaccharide containing sialic acid	Negatively charged neuronal membrane glycolipid.
Cardiolipin	Diphosphatidylglycerol	Inner mitochondrial membrane organization.
Dipalmitoylphosphatidylcholine	Saturated phosphatidylcholine species	Important pulmonary surfactant lipid.

Phospholipase A₂ releases the fatty acid at glycerophospholipid position 2, often arachidonate. Phospholipase C cleaves PIP₂ into DAG and IP₃: DAG remains membrane associated and activates protein kinase C, while IP₃ releases stored intracellular calcium. A GPI anchor attaches selected proteins to the external membrane surface. Loss of protective GPI-anchored complement regulators on blood cells contributes to paroxysmal nocturnal hemoglobinuria.

Disease	Enzyme / accumulated substrate	Distinguishing clue
Tay-Sachs	Hexosaminidase A / GM2 ganglioside	Progressive neurodegeneration; classic absence of hepatosplenomegaly.
Gaucher	Glucocerebrosidase / glucocerebroside	Macrophage storage, organ enlargement and bone disease.
Niemann-Pick A/B	Sphingomyelinase / sphingomyelin	Sphingomyelin storage; type C instead concerns intracellular cholesterol trafficking.

Disease	Enzyme / accumulated substrate	Distinguishing clue
Fabry	Alpha-galactosidase A / globotriaosylceramide	X-linked exception; painful neuropathy, angiokeratomas and renal/cardiac disease.
Krabbe	Galactocerebrosidase	Demyelination with toxic psychosine accumulation.
Metachromatic leukodystrophy	Arylsulfatase A / sulfatides	Central and peripheral demyelination.

Arachidonic acid can enter cyclooxygenase pathways to prostaglandins and thromboxanes or lipoxygenase pathways to leukotrienes. PGI₂ generally supports vasodilation and inhibits platelet aggregation; TXA₂ favors aggregation and vasoconstriction. LTB₄ recruits leukocytes, while cysteinyl leukotrienes promote bronchoconstriction and vascular permeability. Aspirin irreversibly inhibits existing platelet COX-1, but does not block leukotriene synthesis. Its antiplatelet effect persists across the affected platelet lifespan and recovery is mainly through production of new platelets, unlike nucleated endothelium replenishing its enzyme. Do not turn this into the absolute claim that platelets cannot make any protein: they retain translational machinery.

Anti-inflammatory glucocorticoids reduce arachidonate/eicosanoid availability partly through annexin-mediated restraint of phospholipase A₂ and altered gene expression. This differs from aspirin acting at cyclooxygenase. Neither statement means all inflammatory mediators use the same pathway.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.17 pp.374-405

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition - PDF pp. 401-404

Cholesterol, bile acids and steroid synthesis

SOURCE BASIS: book extension

Cholesterol has a steroid nucleus and a C3 hydroxyl group. Esterification of that hydroxyl produces a more hydrophobic storage/transport form. Carbon comes from acetyl-CoA and reducing power from NADPH. Cytosolic/ER cholesterol synthesis proceeds through HMG-CoA, mevalonate, activated isoprenes, squalene and lanosterol. HMG-CoA reductase makes mevalonate and is a key regulated step. It must be distinguished from mitochondrial HMG-CoA synthase/lyase in ketogenesis.

- Low cellular cholesterol allows SREBP processing and increased expression of HMG-CoA reductase and LDL receptors. High sterol availability suppresses this program and promotes reductase degradation.
- AMPK phosphorylation inhibits reductase. Insulin-associated dephosphorylation favors activity. Statins competitively inhibit HMG-CoA reductase, increasing hepatic demand for LDL-derived cholesterol.
- Humans do not completely oxidize the cholesterol ring to CO₂ for energy. Elimination is through free cholesterol in bile and conversion to bile acids. Cholesterol 7- α -hydroxylase is a regulated step in the classical bile-acid pathway.
- Primary bile acids are cholic and chenodeoxycholic acids. Conjugation with glycine or taurine improves detergent function. Intestinal bacteria generate secondary bile acids including deoxycholic and lithocholic acids.

- Efficient ileal reabsorption and portal return create the enterohepatic circulation. Loss of ileal absorption can reduce the bile-salt pool and compromise fat absorption. Cholesterol gallstones form when bile cannot keep cholesterol soluble.

Steroid synthesis begins with cholesterol delivery into mitochondria and side-chain cleavage to pregnenolone. Tissue-specific enzyme expression then directs production toward glucocorticoids, mineralocorticoids or sex steroids. Steroid hormones generally bind intracellular receptors and alter transcription. Deficient steroid enzymes alter both downstream hormone production and upstream precursors, so clinical questions should be reasoned through the blocked step rather than memorized as an isolated name.

Count isoprene carbons: C5 isopentenyl-PP plus its C5 dimethylallyl isomer gives C10 geranyl-PP; addition of another C5 gives C15 farnesyl-PP; two C15 units form C30 squalene. Farnesyl/prenyl groups anchor proteins to membranes; this pathway also supplies nonsterol isoprenoids such as dolichol and ubiquinone. Smith-Lemli-Opitz syndrome is a DHCR7 defect with increased 7-dehydrocholesterol and often low cholesterol. Mevalonate-kinase deficiency blocks phosphorylation downstream of mevalonate and can cause mevalonic aciduria. These pathway markers are established; the rare hematologic distinctions proposed in two supplied questions were not verified and remain withheld.

A net hepatic cholesterol influx greater than efflux means accumulation; it does not by itself identify cholestasis, steatosis or a unique clinical disease. Steroid side-chain cleavage makes pregnenolone from cholesterol. Testosterone-to-dihydrotestosterone conversion requires 5- α -reductase, distinct from aromatase conversion of androgens to estrogens.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.18 pp.406-419,434-442

Mevalonate kinase deficiency with mevalonic aciduria: biochemical case evidence - Accessed 23 September 2026; supplemental clinical verification - <https://pubmed.ncbi.nlm.nih.gov/2850914/>

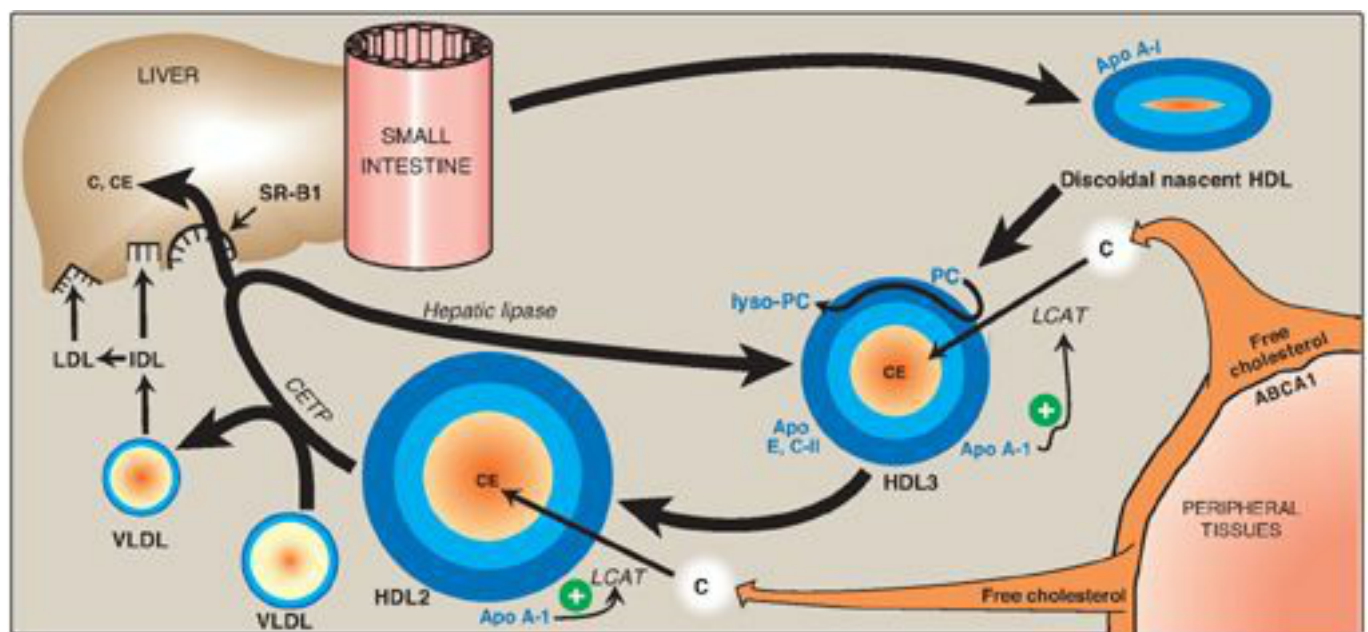
Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition - PDF pp. 410-414; 433-447

Lipoproteins and apoproteins: an integrated transport map

SOURCE BASIS: lecture + book

Particle	Origin and principal job	Recognition / fate
Chylomicron	Intestine; dietary TAG transport	ApoB-48 structure, apoC-II activates LPL, apoE directs remnant uptake.
VLDL	Liver; export of endogenous TAG	ApoB-100; LPL removes TAG to form IDL.
IDL	VLDL remnant	ApoE-dependent liver uptake or further lipolysis to LDL.
LDL	Cholesterol-rich product of VLDL/IDL metabolism	ApoB-100 binds LDL receptor; endocytosis delivers cholesterol.
HDL	Liver/intestine apoA-I, acquiring lipid in plasma	Cholesterol efflux, apoC-II/apoE reservoir, reverse cholesterol transport.

Factor	Function	Do not confuse with
ApoA-I	Activates plasma LCAT	ApoC-II activation of LPL
ApoB-48	Structural protein of chylomicrons	ApoB-100 in VLDL/IDL/LDL
ApoB-100	Structural protein and LDL-receptor ligand	ApoE-mediated remnant recognition
ApoC-II	LPL activator	Pancreatic colipase
ApoE	Remnant receptor ligand	ApoB-48, which does not substitute as the LDL ligand
LCAT	Esterifies cholesterol on HDL using phosphatidylcholine	Intracellular ACAT, which uses acyl-CoA
CETP	Exchanges cholesteryl ester and TAG among HDL and apoB particles	A cholesterol-synthesis enzyme
ABCA1 / SR-B1	Cellular cholesterol efflux to apoA-I / selective hepatic HDL lipid uptake	LDL receptor whole-particle endocytosis



HDL source figure: follow peripheral cholesterol acquisition, LCAT esterification, CETP exchange and delivery to liver. HDL-C concentration alone is not equivalent to all HDL functions.

Identify: ApoA-I; ABCA1; LCAT; CETP; SR-B1

Figure source: Lippincott Biochemistry 6e, PDF p. 432, extracted source diagram

LDL-receptor defects or impaired apoB binding reduce LDL clearance and increase circulating LDL. Macrophages take up modified LDL through scavenger mechanisms that do not provide the same feedback-limited clearance as the LDL receptor, promoting foam cells. Lp(a) is an LDL-like particle with apo(a) attached to apoB-100. Severe chylomicronemia primarily warns of pancreatitis, whereas prolonged exposure to atherogenic apoB particles drives arterial risk.

Pattern	Main particles / mechanism	Exam distinction
Fredrickson I	Chylomicrons; LPL or apoC-II deficiency	Marked TAG, eruptive xanthomas and pancreatitis; tendon xanthomas point more toward LDL excess.
Ila / Iib	LDL alone / LDL plus VLDL	Ila classically LDL-receptor or apoB-100 related; HDL does not have to be low.
III	Remnant particles; impaired apoE-dependent clearance	Both cholesterol and TAG rise; palmar or tuberoeruptive xanthomas suggest it.
IV / V	VLDL / VLDL plus chylomicrons	TAG-predominant patterns, with differing particle contributions.
Tangier / apoB assembly failure	ABCA1-dependent efflux defect / MTP-dependent apoB particle assembly defect	Tangier: extremely low HDL. ApoB-48 assembly failure: fat malabsorption, not the classic LDL-clearance disorder.

Nascent VLDL contains apoB-100; nascent HDL depends on apoA-I. Hepatic lipase helps convert IDL to LDL. Chylomicrons have the highest TAG fraction and LDL the highest cholesterol fraction among the usual choices. HDL carries peripheral cholesterol toward liver; it is not the principal liver-to-periphery cholesterol carrier. Its LCAT reaction uses phosphatidylcholine, unlike ACAT. Post-meal chylomicrons can make plasma milky; persistent fasting chylomicronemia is different.

Insulin resistance commonly combines higher TAG, lower HDL and small dense LDL particles; LDL cholesterol itself need not be high. Hypothyroidism commonly raises LDL by reducing clearance, and HDL changes are variable rather than obligatorily low. Replacing saturated fat with polyunsaturated fat generally lowers LDL, but this is a dietary pattern statement, not an individual treatment prescription.

Sources for this section

Lipid metabolism Esmaeili 1.1.transcript.txt - entire transcript

Lipid metabolism Esmaeili 1.2.transcript.txt - entire transcript

Lipid metabolism1 Esmaeili 1.3.transcript.txt - entire transcript

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.18 pp.420-442

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition - PDF pp. 420-432; 635-650

Protein digestion, amino-acid transport and nitrogen collection

SOURCE BASIS: book extension

Protein digestion starts with acid denaturation and pepsin action in the stomach. Pancreatic proteases are secreted as zymogens. Enteropeptidase initiates trypsin activation at the intestinal surface, and trypsin activates other zymogens. Brush-border and intracellular peptidases complete digestion. Amino acids and small peptides use specific transport systems; transport defects can produce selective urinary losses even when metabolism inside the cell is intact.

Body proteins continually turn over. Ubiquitin tagging directs many selected intracellular proteins to the proteasome, whereas lysosomes degrade endocytosed material and proteins delivered through

autophagy. There is no dedicated storage polymer for surplus amino acids. Their nitrogen must be disposed of while carbon skeletons enter energy metabolism or synthesis.

Reaction/process	Meaning	Key example
Transamination	Transfers an amino group to an alpha-keto acid without releasing free ammonia	ALT: alanine + alpha-ketoglutarate gives pyruvate + glutamate.
AST reaction	Transfers glutamate nitrogen to oxaloacetate	Produces aspartate for the second nitrogen of urea.
Oxidative deamination	Glutamate dehydrogenase releases ammonia and regenerates alpha-ketoglutarate	Mitochondrial reaction links nitrogen collection to urea synthesis.
Glutamine formation	Glutamine synthetase captures ammonia using ATP	Safe nitrogen transport from tissues.
Glutaminase	Releases glutamine amide nitrogen as ammonia	Liver disposal and renal ammonium handling.
Glucose-alanine cycle	Muscle transfers amino nitrogen to pyruvate, exports alanine	Liver uses pyruvate for glucose and nitrogen for urea.

PLP versus NAD(P)+

Aminotransferases require PLP (B6). Glutamate dehydrogenase uses a nicotinamide coenzyme. A transaminase does not consume ATP to remove nitrogen as ammonia; it redistributes nitrogen to another carbon skeleton.

Specify the membrane: many apical amino-acid carriers are sodium coupled, while basolateral release toward portal blood commonly uses facilitated carriers/exchange systems. A question about release into portal blood is not asking how amino acids first enter the brush border. Peptides can enter through proton-coupled transport and then be hydrolyzed. Proelastase, trypsinogen and chymotrypsinogen are zymogens. Glutamine synthetase uses ATP and free ammonia; asparagine synthetase uses ATP and generally glutamine amide nitrogen. Glutamate dehydrogenase can use NAD+ or NADP+; reductive amination commonly uses NADPH.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.19 pp.448-463

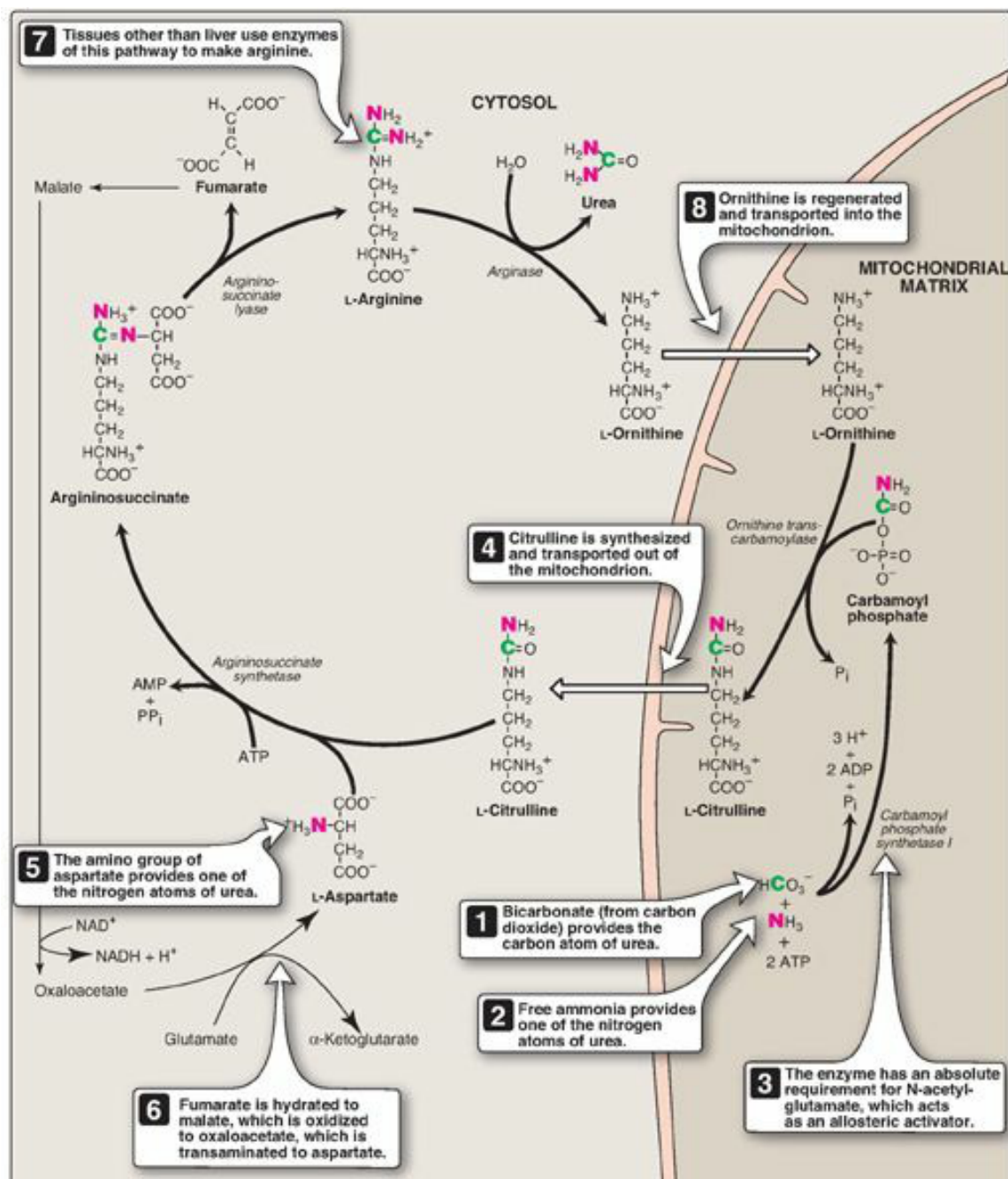
Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition - PDF pp. 448-462

Urea cycle: atom origins, compartments and diagnostic contrasts

SOURCE BASIS: book extension

The liver converts ammonia to urea for renal excretion. Urea receives one nitrogen from free ammonia, one from aspartate, and its carbon from bicarbonate. The first two reactions occur in mitochondrial matrix; later reactions occur in cytosol. Ornithine and citrulline transport joins the compartments.

Enzyme	Reaction	Energy / control
CPS-I	Ammonia + bicarbonate to carbamoyl phosphate	Consumes two ATP; requires N-acetylglutamate.
Ornithine transcarbamylase	Ornithine + carbamoyl phosphate to citrulline	Mitochondrial; citrulline exported.
Argininosuccinate synthetase	Citrulline + aspartate to argininosuccinate	ATP to AMP + PP _i , costing two high-energy bonds.
Argininosuccinate lyase	Argininosuccinate to arginine + fumarate	Fumarate connects to malate/oxaloacetate and aspartate.
Arginase	Arginine + water to ornithine + urea	Ornithine recycled; urea released.



Source urea-cycle figure, showing both compartments and the origin of each nitrogen atom. Follow transport as well as chemistry.

Identify: CPS-I; OTC; Citrulline; Aspartate; Argininosuccinate; Arginine; Ornithine; Fumarate

Figure source: Lippincott Biochemistry 6e, PDF p. 465, extracted source diagram

Three ATP molecules are consumed, but four high-energy phosphate bonds are spent. N-acetylglutamate synthase uses acetyl-CoA and glutamate and is stimulated by arginine. Increased nitrogen intake or catabolism increases cycle demand. Hyperammonemia disrupts brain metabolism and astrocyte glutamine handling, producing neurological dysfunction; no single ammonia level alone identifies the enzyme defect.

Contrast	CPS-I / urea cycle	CPS-II / pyrimidine synthesis
Location	Mitochondria	Cytosol
Nitrogen donor	Free ammonia	Glutamine

Contrast	CPS-I / urea cycle	CPS-II / pyrimidine synthesis
Activator / feedback	N-acetylglutamate required	PRPP activates; UTP inhibits
Defect clue	Hyperammonemia with low citrulline in an early block	Different nucleotide synthesis phenotype

OTC versus CPS-I deficiency

OTC deficiency is X-linked and can divert excess carbamoyl phosphate into pyrimidine synthesis, raising urinary orotic acid. CPS-I deficiency blocks carbamoyl phosphate formation and typically lacks that orotic-acid rise. Both can cause hyperammonemia; primary UMP-synthase deficiency causes orotic aciduria without a primary urea-cycle block.

OTC deficiency is the common inherited urea-cycle disorder and is X-linked, whereas CPS-I deficiency is a different early block. Arginase deficiency produces hyperargininemia. The fumarate-to-malate-to-oxaloacetate/aspartate connection joins nitrogen disposal with TCA chemistry. In advanced liver failure, ammonia-to-urea disposal falls; a low urea value is not automatically evidence of excellent renal filtration.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.19 pp.464-474; Ch.22 pp.551-555

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition - PDF pp. 463-476

Amino-acid carbon skeletons, essentiality and one-carbon metabolism

SOURCE BASIS: book extension

Glucogenic amino acids yield pyruvate or TCA intermediates that can provide net glucose. Ketogenic amino acids yield acetyl-CoA or acetoacetate precursors. Leucine and lysine are exclusively ketogenic. Phenylalanine, tyrosine, tryptophan and isoleucine have both routes. Threonine classification differs among teaching schemes: the inspected 6th edition emphasizes its glucogenic route. Know the specific reaction shown rather than assuming a list copied from another edition is identical.

Entry product	Representative amino acids	Useful link
Pyruvate	Alanine, serine, glycine, cysteine; portions of tryptophan metabolism	Gluconeogenesis through pyruvate carboxylase.
Oxaloacetate	Aspartate, asparagine	Direct connection to AST and urea nitrogen.
Alpha-ketoglutarate	Glutamate, glutamine, proline, arginine, histidine	Nitrogen collection and TCA replenishment.
Succinyl-CoA	Methionine, valine, isoleucine, threonine pathways	Propionyl-CoA, biotin and B12 link.
Fumarate	Phenylalanine and tyrosine	Their other carbon fragments are ketogenic.

Entry product	Representative amino acids	Useful link
Acetyl-CoA/acetoacetate	Leucine, lysine and ketogenic portions of mixed amino acids	No net conversion of acetyl carbon into glucose.

The essential adult amino acids are histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan and valine. Tyrosine becomes conditionally essential when phenylalanine cannot be hydroxylated. Cysteine depends on an adequate sulfur supply from methionine. Branched-chain amino acids are initially metabolized substantially in peripheral tissues, especially muscle; the liver is not the sole site of all amino-acid catabolism.

Methionine plus ATP forms S-adenosylmethionine, a major methyl donor. After methyl transfer, S-adenosylhomocysteine yields homocysteine. Remethylation to methionine requires methyl-B12 and methyl-THF. Transsulfuration combines homocysteine with serine via B6-dependent cystathionine beta-synthase and ultimately yields cysteine. THF carries one-carbon units at different oxidation states for purines, thymidylate and selected amino-acid reactions. Biotin carries CO₂ in carboxylation, not the same one-carbon pool.

Carbon classification in the inspected 6e	Amino acids	Reasoning
Exclusively ketogenic	Leucine, lysine	Yield acetyl/acetoacetyl carbon without a net glucogenic product.
Both glucogenic and ketogenic	Isoleucine, phenylalanine, tyrosine, tryptophan	Generate a glucogenic entry intermediate plus acetyl-CoA or acetoacetate.
Glucogenic in this 6e table	Alanine, arginine, asparagine, aspartate, cysteine, glutamate, glutamine, glycine, histidine, methionine, proline, serine, threonine, valine	Produce pyruvate or a net TCA intermediate.

Threonine classification depends on the represented route

The inspected 6e Figure 20.2 lists threonine as glucogenic and emphasizes its alpha-ketobutyrate/propionyl-CoA route. Some teaching lists classify it as both by including acetyl-CoA-producing routes; the source summary Figure 20.14 itself depicts such an alternate route faintly. For an exam, identify the route or the named source rather than presenting one list as a universal contradiction-free rule.

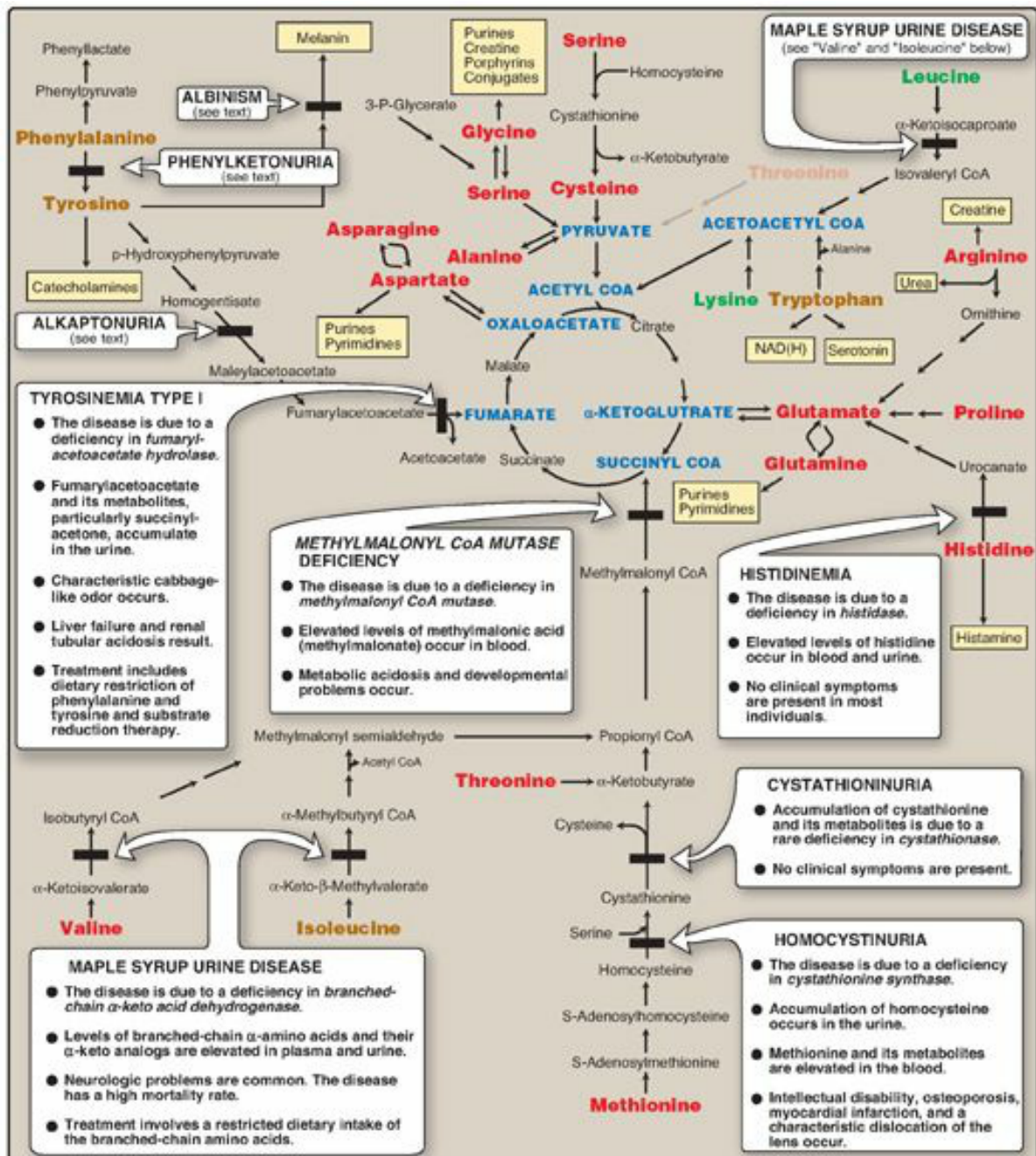


Figure 20.14 from the local book maps amino-acid carbon entry, special products and inherited blocks. The adjacent table states the edition-specific threonine convention; text below explains the tyrosinemia type I box.

Identify: Pyruvate; Oxaloacetate; Alpha-ketoglutarate; Succinyl-CoA; Fumarate; Acetyl-CoA; Acetoacetyl-CoA

Figure source: Lippincott Biochemistry 6e, PDF p. 493, extracted source diagram

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.20 pp.477-493,501-502

Amino-acid disorders: reason from the blocked reaction

SOURCE BASIS: book extension

Condition	Block	Reasoning pattern
Classic PKU	Phenylalanine hydroxylase	High phenylalanine, low tyrosine; neurological toxicity and reduced pigment.
BH4-related hyperphenylalaninemia	BH4 synthesis or regeneration	Also affects tyrosine and tryptophan hydroxylases, so neurotransmitter deficits accompany elevated phenylalanine.
Maple syrup urine disease	Branched-chain alpha-ketoacid dehydrogenase	Leucine, isoleucine and valine-derived metabolites accumulate; infantile encephalopathy and characteristic odor.
Classic homocystinuria	Cystathionine beta-synthase	High homocysteine and often methionine; thrombosis, lens displacement and skeletal phenotype.
B12 or remethylation defect	Homocysteine to methionine impaired	Homocysteine rises but methionine need not rise; methylmalonate helps identify B12 involvement.
Alkaptonuria	Homogentisate oxidase	Urine darkens on standing, ochronosis and large-joint disease.
Tyrosinase-related albinism	Melanin synthesis	Pigment/visual phenotype without the phenylalanine accumulation of PKU.
Methylmalonyl-CoA mutase or B12 defect	Methylmalonyl-CoA to succinyl-CoA	Methylmalonic acid accumulation and metabolic dysfunction.

Clinical dietary management of inborn errors must preserve essential nutrient requirements while limiting toxic accumulation; it is not simply complete removal of all protein. Maternal hyperphenylalaninemia can harm fetal development even if the fetus does not have PKU. For examination reasoning, separate substrate accumulation, product deficiency and secondary pathway diversion. These summaries are learning material, not individual treatment protocols.

Tyrosinemia type I is deficiency of fumarylacetoacetate hydrolase, the terminal tyrosine-degradation step. Fumarylacetoacetate and toxic metabolites including succinylacetone accumulate; hepatic failure and renal tubular dysfunction are characteristic. Succinylacetone is a particularly useful biochemical discriminator from PKU or alkaptonuria. This mechanism is shown in the source Figure 20.14 at PDF p.493. Histidase deficiency is another named block in that figure and often has no major clinical phenotype; do not assume that every elevated amino acid causes the same severe neonatal syndrome.

Hartnup disease affects neutral amino-acid transport in intestine and kidney, including tryptophan and phenylalanine. Loss of tryptophan limits niacin synthesis and can produce pellagra-like dermatitis and neurologic features. It differs from cystinuria, which affects cystine and dibasic amino acids. Tyrosinemia II is tyrosine aminotransferase deficiency: corneal pseudodendritic lesions and palmoplantar skin lesions distinguish it from type I liver/renal disease; eye lesions may resemble

herpes keratitis. PKU causes phenylalanine accumulation and tyrosine deficiency, not a typical cirrhotic-liver presentation.

BH4 is synthesized from GTP and is required by phenylalanine, tyrosine and tryptophan hydroxylases and nitric-oxide synthase. Dopamine beta-hydroxylase instead requires copper and ascorbate. Arginine is conditionally essential during growth or some illnesses. In classic CBS-deficient homocystinuria, homocysteine and methionine often rise; remethylation defects differ. MSUD neurotoxicity involves leucine and branched-chain ketoacids. The maple odor is associated with sotoson but odor alone cannot establish the defect or justify unsupported EXCEPT lists.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.20 pp.494-504

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition - PDF pp. 477-505; 521-527

GeneReviews: Tyrosinemia Type II - Accessed 23 September 2026; supplemental clinical verification - <https://www.ncbi.nlm.nih.gov/books/NBK608431/>

Heme synthesis, porphyrias and bilirubin handling

SOURCE BASIS: book extension

Heme synthesis begins in mitochondria when glycine and succinyl-CoA form ALA through PLP-dependent ALA synthase. Several intermediate steps occur in cytosol before the pathway returns to mitochondria. ALA dehydratase forms porphobilinogen, successive condensation/cyclization steps build the porphyrin ring, and ferrochelatase inserts ferrous iron into protoporphyrin IX. Hepatic ALAS1 responds to heme feedback; erythroid ALAS2 is strongly linked to iron availability. Lead inhibits ALA dehydratase and ferrochelatase.

Porphyrias differ by the accumulating intermediate. Early precursor accumulation can cause neurovisceral attacks, whereas later porphyrin accumulation can create photosensitivity. The name of the enzyme alone is insufficient: identify whether precursors or light-reactive porphyrins accumulate. Heme breakdown by heme oxygenase produces biliverdin, carbon monoxide and iron. Biliverdin reductase forms bilirubin. Unconjugated bilirubin travels bound to albumin; hepatic UDP-glucuronosyltransferase conjugates it, enabling biliary secretion.

Pattern	Serum/urine reasoning	Mechanism
Hemolysis	Predominantly unconjugated bilirubin; no unconjugated bilirubin in urine	Increased heme load exceeds conjugation capacity; albumin-bound pigment is not filtered normally.
Obstruction	Conjugated bilirubin can appear in urine; reduced pigment reaching bowel	Biliary outflow impaired; pale stools can occur.
Hepatocellular injury	Mixed patterns possible	Both conjugation and secretion can be impaired.
Neonatal low conjugation capacity	Unconjugated hyperbilirubinemia	Developmental enzyme limitation must be distinguished from obstruction.

Two different conjugation systems

Bilirubin conjugation uses glucuronic acid. Bile acids are conjugated with glycine or taurine. Cholesterol esterification adds a fatty acid. These are distinct reactions involving different substrates and enzymes.

Porphyria	Deficient enzyme	Useful accumulation/pattern contrast
Acute intermittent porphyria	Hydroxymethylbilane synthase / porphobilinogen deaminase	ALA and PBG rise; neurovisceral attacks without the characteristic porphyrin photosensitivity.
Porphyria cutanea tarda	Uroporphyrinogen decarboxylase	Uroporphyrin-related accumulation; predominantly cutaneous photosensitivity and liver/iron-related associations.
Hereditary coproporphyria	Coproporphyrinogen oxidase	Coproporphyrin-related accumulation; acute attacks may coexist with photosensitivity.
Variegate porphyria	Protoporphyrinogen oxidase	Protoporphyrinogen and upstream metabolites accumulate; acute and cutaneous manifestations can coexist.
Erythropoietic protoporphyria	Ferrochelatase in the classic book pattern	Protoporphyrin accumulation in erythrocytes/plasma and hepatobiliary consequences; photosensitivity.
Congenital erythropoietic porphyria	Uroporphyrinogen III synthase	Type I porphyrin production/accumulation; early photosensitivity.

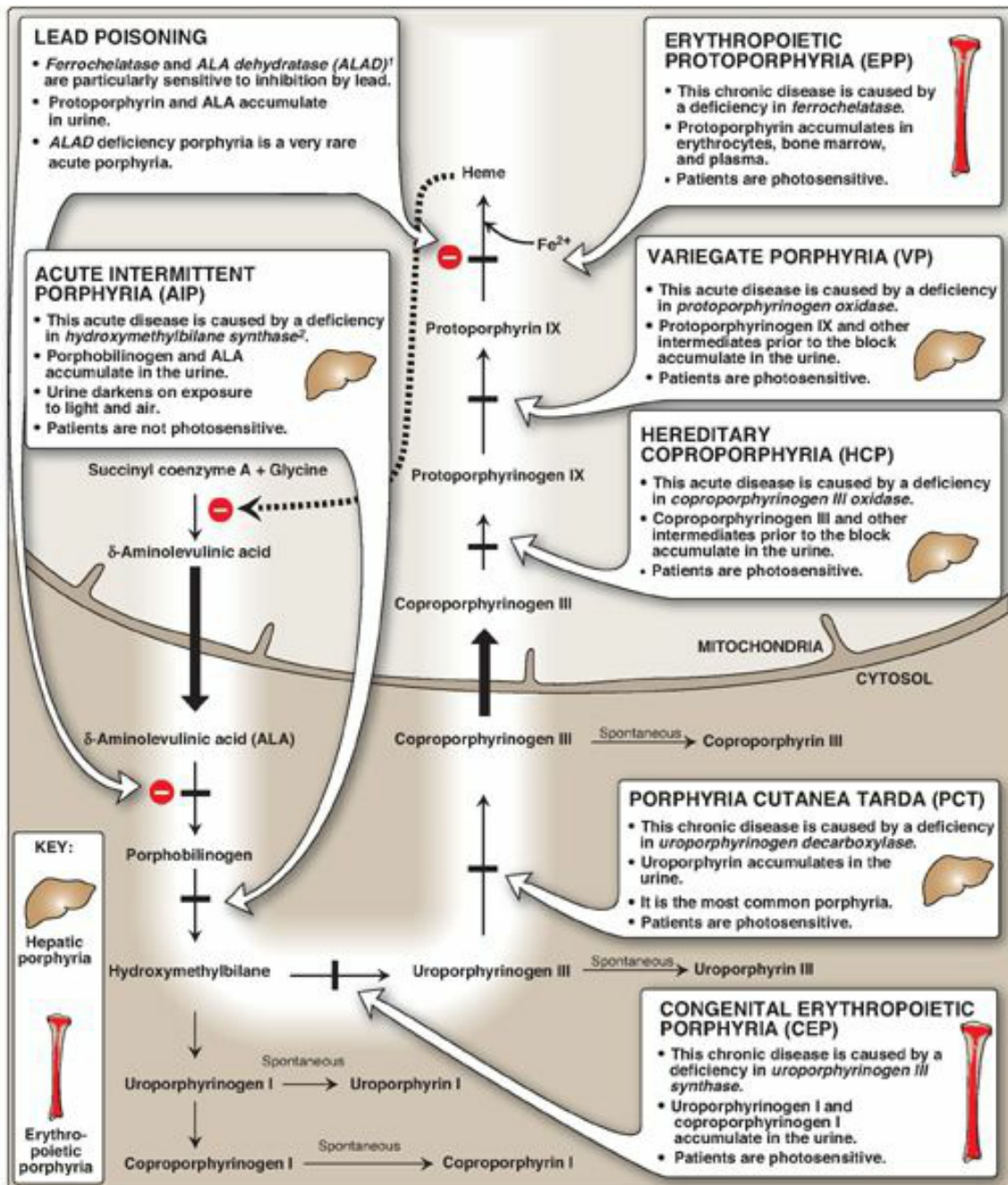


Figure 21.8, local Lippincott PDF p.515 (caption begins p.514): mitochondrial/cytosolic heme pathway and porphyria blocks. Distinguish precursor accumulation from mature heme deficiency.

Identify: ALA synthase; ALA dehydratase; Hydroxymethylbilane synthase; Uroporphyrinogen III synthase; Uroporphyrinogen decarboxylase; Coproporphyrinogen oxidase; Protoporphyrinogen oxidase; Ferrochelatase

Figure source: Lippincott Biochemistry 6e, PDF p. 515, extracted source diagram

A pathway block does not mean all tests rise equally

Acute precursor-producing disorders and photosensitizing porphyrin accumulation are related but distinct patterns. Lead inhibits ALA dehydratase and ferrochelatase; it is an acquired toxic block, not automatically one inherited porphyria. ALAS2 deficiency causes sideroblastic anemia rather than an ordinary porphyria phenotype. This review does not turn the older book treatment paragraph into current management advice.

Disorder	Mechanism	Bilirubin pattern
Gilbert	Partial reduction in UGT1A1 activity	Mild intermittent unconjugated rise; no bilirubinuria. A fixed decrease in urine urobilinogen is not diagnostic.
Crigler-Najjar I / II	Absent / markedly reduced conjugating activity	Severe / less severe unconjugated hyperbilirubinemia; reduced pigment delivery to gut in severe conjugation failure.
Dubin-Johnson	Canalicular conjugated-bilirubin export defect	Conjugated hyperbilirubinemia; distinguish it from UGT1A1 failure.
Neonatal physiologic pattern	High bilirubin load and immature uptake/conjugation	Predominantly unconjugated, not direct hyperbilirubinemia.
Hemolysis / complete obstruction	More heme delivery / little bilirubin reaches bowel	Hemolysis: increased urinary urobilinogen. Complete obstruction: little or absent urobilinogen, dark bilirubin-containing urine and pale stools.

Heme oxygenase opens the ring using oxygen and NADPH; biliverdin reductase uses reducing equivalents to form bilirubin. Gut bacteria form urobilinogen from bilirubin; most becomes fecal stercobilin, while some is reabsorbed and a small amount is excreted/oxidized to urinary urobilin. ALAS1 is the broadly expressed hepatic/nonerythroid regulatory isoform; ALAS2 is erythroid, not ubiquitous. CEP often presents early and severely; PCT is the common predominantly acquired cutaneous form. Increased urine ALA supports a lead-related pathway block, but blood lead measurement establishes exposure; a paper offering only urinary ALA does not make it the definitive modern diagnostic test.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.21 pp.506-529

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition - PDF pp. 507-520

Amino acids as precursors of signaling and storage molecules

SOURCE BASIS: book extension

Precursor	Product sequence	Cofactor / significance
Tyrosine	DOPA, dopamine, norepinephrine, epinephrine	Tyrosine hydroxylase uses BH ₄ ; decarboxylation uses PLP; epinephrine methylation uses SAM.
Tryptophan	5-hydroxytryptophan, serotonin, melatonin	Hydroxylation uses BH ₄ and decarboxylation uses PLP.
Histidine	Histamine	PLP-dependent decarboxylation.
Glutamate	GABA	PLP-dependent glutamate decarboxylase.
Arginine	Nitric oxide and citrulline	NOS uses oxygen and reducing power; local vascular and neural signaling.
Glycine + arginine + methyl group from methionine	Creatine, then phosphocreatine	Creatine kinase buffers ATP demand; spontaneous creatinine formation links muscle pool to renal lab measurements.
Glycine, cysteine, glutamate	Glutathione	Antioxidant tripeptide; reduced and oxidized pools must be distinguished.
Tyrosine	Melanin	Copper-dependent tyrosinase, not phenylalanine hydroxylase.

Amino-acid chemistry therefore explains more than fuel oxidation. A deficiency of a shared cofactor can affect neurotransmission, heme synthesis and transamination together. SAM methyl-transfer reactions and THF/B12-dependent remethylation link one-carbon metabolism to these specialized products.

Creatine synthesis uses glycine and arginine plus a methyl group donated by methionine through SAM. Kidney and liver are major contributors, but brain can also synthesize creatine.

Creatine/phosphocreatine spontaneously cyclize to creatinine; do not mistake creatinine for the ATP-buffering substrate phosphocreatine. Nitric oxide activates soluble guanylyl cyclase and increases cGMP, favoring smooth-muscle relaxation rather than contraction.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.21 pp.520-529

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition - PDF pp. 521-527

Purines, pyrimidines, salvage and DNA precursor supply

SOURCE BASIS: book extension

A nucleoside contains a base and sugar; a nucleotide adds phosphate. Purines are built on a pre-existing ribose-phosphate scaffold supplied by PRPP. The committed purine step is glutamine-PRPP amidotransferase, stimulated by substrate availability and inhibited by end products. IMP branches toward AMP or GMP, with reciprocal use of GTP and ATP helping balance their production. One-carbon units from formyl-THF participate in purine-ring construction.

Purine salvage attaches bases to PRPP through APRT or HGPRT, conserving energy. Purine degradation ends in uric acid in humans. Xanthine oxidase acts in the hypoxanthine-to-xanthine-to-urate sequence. HGPRT deficiency reduces salvage and removes feedback restraint on de novo synthesis; severe deficiency causes Lesch-Nyhan syndrome. Hyperuricemia can result from excess production or impaired excretion and is not by itself identical to symptomatic gout.

Pyrimidine rings are built before attachment to ribose phosphate. Cytosolic CPS-II supplies carbamoyl phosphate using glutamine. The ring proceeds through orotate, which becomes OMP and UMP through UMP-synthase activities. A primary UMP-synthase defect causes orotic aciduria with impaired nucleotide supply, unlike OTC deficiency where orotate rises secondarily to a urea-cycle block.

Enzyme	Function	High-yield association
Ribonucleotide reductase	Converts ribonucleoside diphosphates to deoxy forms	Overall inhibition by dATP; hydroxyurea target.
Thymidylate synthase	dUMP to dTMP using methylene-THF	5-fluorouracil-derived inhibitor; folate becomes DHF.
Dihydrofolate reductase	Regenerates THF from DHF using NADPH	Methotrexate target; needed to sustain nucleotide synthesis.
Adenosine deaminase	Purine breakdown	Deficiency raises deoxyadenosine/dATP, impairing lymphocyte DNA synthesis and causing SCID.
CPS-II	First regulated pyrimidine step	UTP inhibits; PRPP activates; cytosolic and glutamine dependent.

Ring / compound	Origins or identity	Common trap
Purine N1; N3/N9; C4/C5/N7	Aspartate; glutamine; glycine	Free ammonia does not replace glutamine amide nitrogen in this pathway.
Purine C2/C8; C6	N10-formyl-THF; CO ₂ /bicarbonate	Glycine participates in purines, not pyrimidine-ring synthesis.
Pyrimidine N1/C4/C5/C6; N3; C2	Aspartate; glutamine; CO ₂ /bicarbonate	Identify the depicted ring before interpreting numbered source labels.

Ring / compound	Origins or identity	Common trap
IMP / OMP / UMP	First completed purine nucleotide / first ribose-linked pyrimidine nucleotide / product after OMP decarboxylation	OMP and UMP answer different questions. Inosine is hypoxanthine plus ribose, without phosphate.
Pyrimidine degradation	Uracil/cytosine to beta-alanine; thymine to beta-aminoisobutyrate	Urate is the human purine endpoint, not a pyrimidine endpoint.

Thioredoxin supplies reducing equivalents to ribonucleotide reductase and is regenerated through NADPH-dependent thioredoxin reductase. Mycophenolic acid inhibits IMP dehydrogenase and lowers GMP synthesis. Allopurinol/oxypurinol inhibit xanthine oxidase, lowering urate and increasing upstream hypoxanthine/xanthine; salvage can reduce de novo flux. Methotrexate blocks DHFR and indirectly limits dTMP and purine synthesis. The 5-fluorouracil metabolite inhibits thymidylate synthase, not an enzyme called thymidine synthase. Sulfonamides block bacterial dihydropteroate synthase upstream of folate-dependent nucleotide production, not human CPS-II directly.

Tophi are deposits of monosodium urate in gout; elevated serum urate alone does not prove gout. High cell turnover, impaired renal excretion and thiazides can increase urate. Fanconi proximal-tubular dysfunction can instead waste urate. Do not generalize an EXCEPT option naming non-thiazide antihypertensives to every drug in that broad group. ADA deficiency raises dATP, suppresses DNA precursor production and reduces functional lymphocyte populations; not every SCID is caused by ADA deficiency.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.22 pp.532-559

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition - PDF pp. 536-555

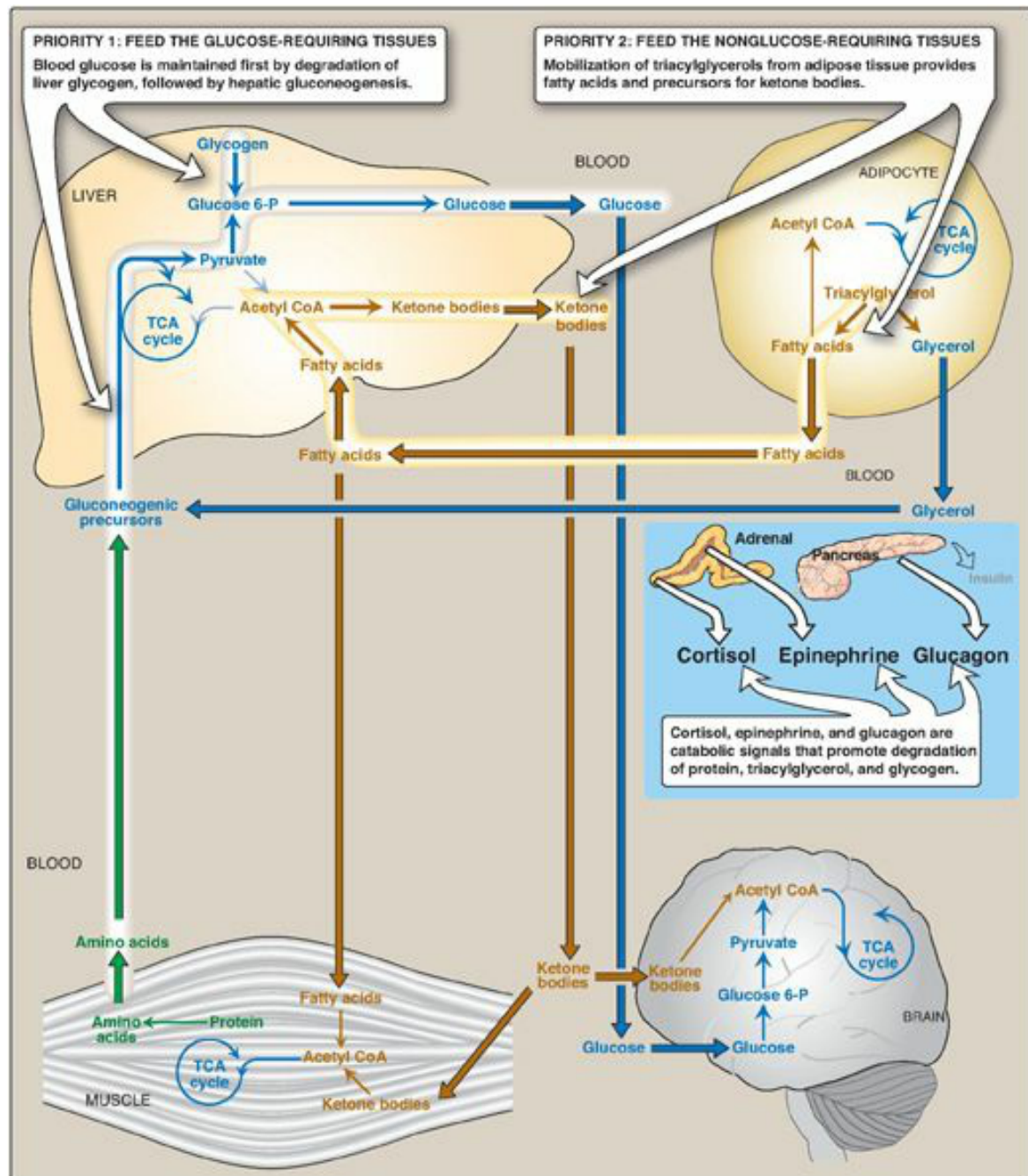
Insulin, glucagon and the fed-fast cycle

SOURCE BASIS: book extension

Pancreatic beta cells process proinsulin to insulin and C-peptide. Glucose metabolism raises ATP, closes ATP-sensitive K⁺ channels, depolarizes the cell and permits calcium entry that triggers secretion. Insulin promotes glucose uptake in muscle/adipose, glycogen synthesis, lipogenesis and protein synthesis. Glucagon from alpha cells primarily coordinates hepatic glucose output and ketogenesis. Amino acids can stimulate both hormones after a protein-containing meal, helping avoid a fall in blood glucose as amino acids are taken up.

Tissue	Fed state	Fasting / prolonged fasting
Liver	Glycogen replenishment, glycolysis, fatty-acid/TAG synthesis and VLDL export	Early glycogenolysis, then increasing gluconeogenesis; fatty-acid oxidation supplies energy and ketones.
Adipose	GLUT4 uptake, TAG synthesis and storage	TAG lipolysis exports fatty acids and glycerol.
Skeletal muscle	Glucose uptake, glycogen and protein synthesis	Fatty acids and ketones support energy; early protein release supplies gluconeogenic amino acids.

Tissue	Fed state	Fasting / prolonged fasting
Brain	Predominantly glucose	Increasing ketone use over prolonged fasting reduces glucose requirement.
RBC	Glucose to lactate throughout	Still needs glucose and PPP NADPH; cannot use fatty acids or ketones.
Kidney cortex	Oxidative metabolism and routine acid-base handling	Increased gluconeogenesis and ammonium excretion in prolonged fasting.



Book concept figure linking tissues during nutrient deprivation. Use the table to distinguish early fasting from the later adaptation that spares protein.

Identify: Liver; Adipose tissue; Muscle; Brain; Glucose; Fatty acids; Ketones

Figure source: Lippincott Biochemistry 6e, PDF p. 609, extracted source diagram

The purpose of adaptation is not to stop glucose production, but to reserve glucose for tissues that require it while making alternative fuel available. As the brain uses more ketones, muscle protein breakdown can diminish. In exercise, ATP demand rises rapidly, calcium helps coordinate glycogen breakdown and TCA oxidation, and muscle contraction can increase glucose uptake independently of insulin.

Mature insulin contains 51 residues in two chains and THREE disulfide bridges: two between chains and one within the A chain. A question asking total bridges with only two offered is flawed. Endogenous insulin and C-peptide are released together; C-peptide is low in severe beta-cell failure and does not accompany injected insulin. Diazoxide opens beta-cell KATP channels and suppresses secretion; drug-induced diabetes belongs to other specific types. Catecholamine alpha-2 signaling inhibits insulin release during stress. Glucagonoma favors glycogenolysis/gluconeogenesis and inhibits glycogenesis/lipogenesis.

Distinguish early fasting from prolonged starvation: muscle can use ketones but progressively favors fatty acids, conserving ketones for brain. Fed insulin activates adipose LPL and suppresses intracellular HSL-mediated lipolysis. The first delivers lipid from circulating particles; the second mobilizes stored fat. High carbohydrate supply tends to support hepatic VLDL production; prolonged fasting is not equivalent to an uncontrolled diabetic state.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Chs.23-24 pp.560-613

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition - PDF pp. 562-582; 597-613

Diabetes and metabolic perturbations: causal review

SOURCE BASIS: book extension

Type 1 diabetes typically involves profound insulin deficiency. Hepatic glucose output is insufficiently suppressed while peripheral insulin-dependent glucose use falls. Adipose lipolysis delivers fatty acids to liver, promoting ketogenesis. Type 2 diabetes combines insulin resistance with inadequate beta-cell compensation; residual insulin often restrains ketogenesis, but ketoacidosis is not impossible. These categories describe mechanisms, not an inflexible rule based only on age or body size.

- Hyperglycemia can cause osmotic diuresis and water/electrolyte loss. Serum potassium may be normal or high in ketoacidosis despite total-body potassium depletion, because distribution and loss are separate variables.
- Chronic glucose excess promotes nonenzymatic glycation and oxidative/inflammatory injury. HbA1c reflects prior exposure but is also affected by RBC lifespan and hemoglobin-related factors.
- Aldose-reductase flux can consume NADPH and accumulate sorbitol in susceptible tissues. This is one mechanistic link between hyperglycemia and cellular stress, not the sole explanation of every diabetic complication.
- Visceral adipose dysfunction, excess fatty-acid delivery and altered adipokines contribute to insulin resistance. Leptin signals energy stores, whereas ghrelin promotes hunger; common obesity is not generally simple absence of leptin.
- High NADH from ethanol favors lactate formation and TAG accumulation and suppresses gluconeogenesis. Combine nutritional state, glycogen reserve and redox effects when solving an alcohol-metabolism vignette.

Check yourself: Why can fasting liver oxidize fatty acids while simultaneously making glucose?

Fatty-acid oxidation supplies the ATP and reducing power required for gluconeogenesis. Acetyl-CoA activates pyruvate carboxylase and inhibits PDH, directing pyruvate toward oxaloacetate rather than oxidation. Low malonyl-CoA permits CPT-I-mediated fatty-acid entry. The pathways are coordinated rather than contradictory.

Hyperosmolar hyperglycemic state is more typical of type 2 diabetes: profound hyperglycemia and water loss produce hyperosmolality, with relatively restrained ketogenesis. DKA is especially associated with severe insulin deficiency, but overlap exists. Metabolic (Reaven) syndrome combines central adiposity, insulin resistance, high TAG/low HDL, raised pressure and dysglycemia, increasing cardiovascular risk. Acute illness and assay timing matter; neither HHS nor DKA should be diagnosed from a memorized demographic rule alone.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Chs.23,25,26 pp.575-579,614-650

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition - PDF pp. 614-633

Vitamin and cofactor integration for metabolism

SOURCE BASIS: book extension

Vitamin/cofactor	Reactions to recognize	Deficiency / distinction
B1, TPP	PDH, alpha-KGDH, BCKD and transketolase	Defects link energy failure to neurological/cardiac disease; transketolase is in PPP.
B2, FAD/FMN	Flavoprotein redox reactions, respiratory chain	Not the cofactor for methylmalonyl-CoA mutase.
B3, NAD/NADP	Oxidation/reduction and reductive synthesis	Pellagra pattern: dermatitis, diarrhea and neuropsychiatric dysfunction.
B5, CoA / phosphopantetheine	Acyl transfer and fatty-acid synthase ACP domain	Connects acetyl-CoA, succinyl-CoA and fatty-acyl transport.
B6, PLP	Transamination, selected decarboxylation, ALA synthase, glycogen phosphorylase	Amino-acid, heme and glycogen questions can share this cofactor.
Biotin	Pyruvate, acetyl-CoA and propionyl-CoA carboxylases	Carries activated CO ₂ ; deficiency can affect several carboxylases.
Folate, THF	One-carbon transfer in purine and thymidylate synthesis	Deficiency can cause megaloblastic anemia.
B12	Methionine synthase and methylmalonyl-CoA mutase	Deficiency can raise homocysteine and methylmalonate and cause neurological dysfunction.
C, ascorbate	Collagen hydroxylation and iron handling	Scurvy reflects defective connective tissue.
A, retinal / retinoic acid	Vision / gene regulation and epithelial differentiation	Night blindness or epithelial changes; retinoic acid cannot restore retinal for vision.
D, calcitriol	Calcium/phosphate handling and transcription	Liver 25-hydroxylation then kidney 1-alpha-hydroxylation; deficiency impairs mineralization.
E, tocopherols	Lipid membrane antioxidant protection	Fat-malabsorption states can cause deficiency with neurological/hemolytic manifestations.
K	Gamma-carboxylation of clotting proteins and proteins C/S	Required for calcium-binding Gla residues; impaired vitamin-K recycling is a distinct mechanism.

B12 absorption requires release from food, transfer from binding proteins to intrinsic factor, and ileal uptake. Folate can improve the anemia of B12 deficiency without correcting its neurological injury, so the two deficiencies must be distinguished. Fat-soluble vitamins travel with dietary lipid and are vulnerable to cholestasis, pancreatic insufficiency and chylomicron transport defects. This review

intentionally omits supplement doses: the local PDF has OCR unit artifacts and is not a current prescribing source.

Do not confuse deficiency markers

Both folate and B12 deficiency can increase homocysteine. Increased methylmalonate supports B12-related dysfunction rather than isolated folate deficiency. The local book summary contains a B2/B12 text defect for methylmalonyl-CoA mutase; the detailed reaction at PDF p.361 confirms B12.

B12 transport stage	Required event	Failure point
Stomach	Food-bound B12 is released by gastric conditions and binds R-protein/haptocorrin	Reduced gastric acid impairs food-cobalamin release; do not equate all malabsorption with autoimmunity.
Duodenum	Pancreatic proteases degrade R-protein, allowing B12 to bind intrinsic factor	Pancreatic dysfunction can impair this transfer.
Ileum	Intrinsic-factor-B12 binds specific mucosal receptors and is taken up	Ileal disease/resection or loss of IF disrupts receptor-mediated uptake.
Blood and tissues	B12 travels on transcobalamin and reaches tissues; major stores are hepatic	Transport defects differ from dietary absence.
Enterohepatic recycling	Biliary B12 can be efficiently reclaimed in ileum	Large stores and recycling explain why dietary deficiency may take years.

Pernicious anemia specifically reflects autoimmune intrinsic-factor/parietal-cell-related failure, not every cause of B12 malabsorption. In methionine-synthase failure, folate becomes functionally trapped as methyl-THF, depleting forms required for purine/thymidylate synthesis. This links B12 deficiency to megaloblastosis while the second B12-dependent reaction explains increased methylmalonate.

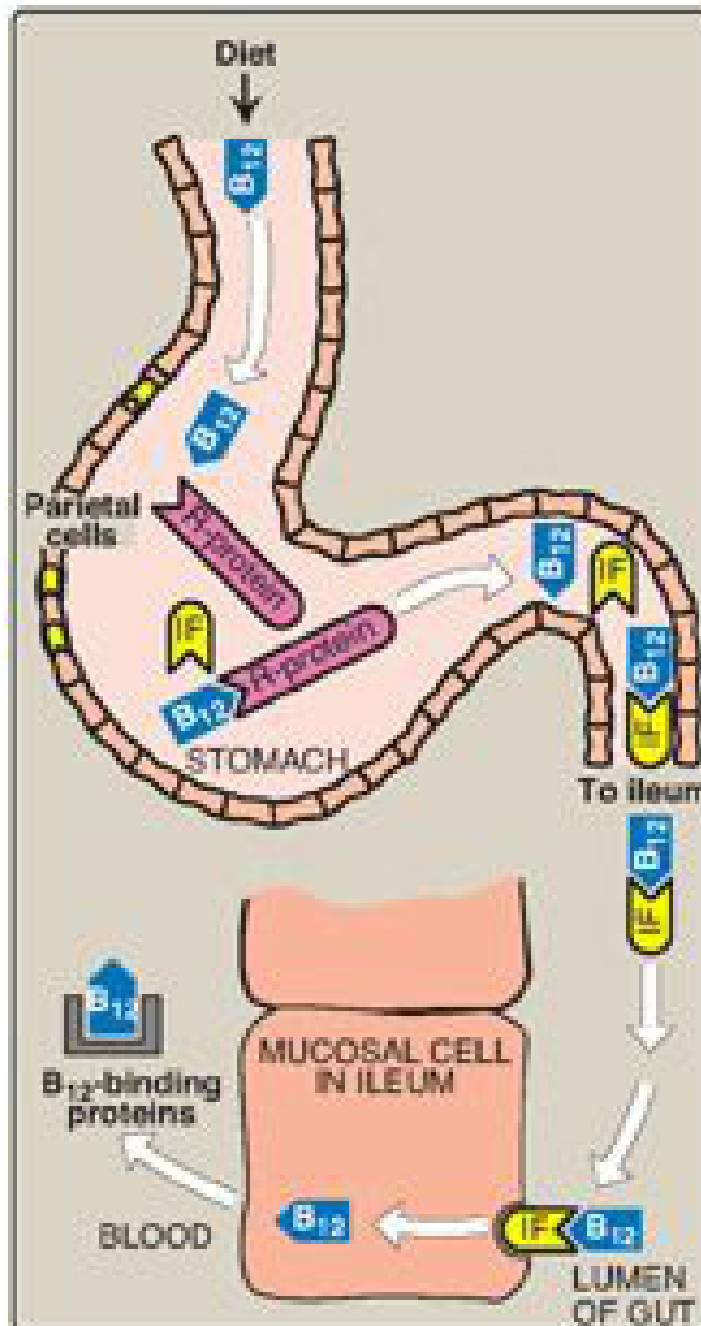


Figure 28.7, local PDF p.688 (caption begins p.687): follow food B₁₂, gastric binding proteins, pancreatic release, intrinsic-factor binding and ileal uptake.

Identify: Stomach; R-protein; Intrinsic factor; Pancreas; Ileum; Blood transport

Figure source: Lippincott Biochemistry 6e, PDF p. 688, extracted source diagram

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.28 pp.681-720; Ch.20 pp.483-489

Nutrition, nitrogen balance and final integration checkpoints

SOURCE BASIS: book extension

Food energy is conventionally approximated as 4 kcal/g for carbohydrate or protein and 9 kcal/g for fat. Energy expenditure includes resting metabolism, activity and the thermic effect of food. These are population-level accounting values rather than a promise that every molecule yields an identical quantity of useful ATP. Protein quality depends on essential-amino-acid supply and digestibility. Combining complementary food sources can improve the overall amino-acid pattern.

Positive nitrogen balance accompanies net tissue growth or recovery. Negative balance occurs when losses exceed intake during inadequate protein supply or catabolic stress. Protein-energy malnutrition can combine energy and protein deficits in different proportions. Edema in a protein-deficient phenotype relates partly to reduced plasma protein and altered fluid distribution, but real malnutrition is more complex than a single albumin value.

Question to ask of any pathway	A complete answer should name
Where?	Tissue and subcellular compartment, including required membrane transport.
What goes in and out?	Carbon count, nitrogen source, energy and reducing-cofactor balance.
What controls flux?	Regulated enzyme, allosteric signals, hormonal modification and tissue exception.
What happens if blocked?	Accumulated substrate, missing product, compensatory pathway and clinical consequence.
What is the source confidence?	Lecture-supported versus book-only extension, plus any edition or transcript discrepancy.

Check yourself: A question gives fasting hypoglycemia. How do ketones refine the metabolic differential?

High ketones indicate that fat mobilization and hepatic ketogenesis are active, as in ordinary fasting or some carbohydrate-pathway disorders. Inappropriately low ketones suggest impaired fatty-acid oxidation/transport or inappropriate insulin effect. Interpret age, fasting duration, lactate, liver/muscle findings and other metabolites before assigning a specific disorder.

Check yourself: An amino acid contributes to both urea and glucose. Must its nitrogen and carbon follow the same route?

No. Transamination separates nitrogen handling from the fate of the carbon skeleton. Nitrogen can reach glutamate, ammonia or aspartate and then urea, while the alpha-ketoacid enters gluconeogenesis or oxidation. This separation is the organizing principle of amino-acid catabolism.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition (2014) - Ch.27 pp.651-680

Diagnostic enzymes, isoenzymes and cardiac/pancreatic markers

SOURCE BASIS: past-paper gap audit + book/reference extension

Separate physiological plasma enzymes from leakage markers. Coagulation enzymes and circulating/endothelial lipoprotein-lipase activity have physiological functions in blood. ALT, AST, LDH and CK ordinarily act inside cells; increased circulating activity can reflect injury. This older functional/nonfunctional terminology describes location of normal action, not whether an enzyme is biologically useful. Isoenzymes catalyze the same reaction but differ in subunits, properties or tissue distribution.

Marker	Main exam association	Qualification
CK / CK-MM / CK-MB / CK-BB	Muscle injury/rhabdomyolysis / skeletal muscle / cardiac-enriched / brain-enriched	CK-MB is not absolutely confined to heart. ALP is not a usual muscle-injury marker.
LDH and the flipped ratio	LDH-1 greater than LDH-2: historical myocardial-injury pattern	RBC hemolysis also raises LDH; sample hemolysis can produce misleading results. Reticulocytes, bilirubin and other evidence help distinguish true hemolysis.
Cardiac troponin I/T	Myocardial injury, including NSTEMI	Preferred contemporary markers; a rise/fall and ischemic context distinguish MI from other myocardial injury.
Myoglobin	Early release from damaged muscle	Older papers call it an early sensitive marker; it is not cardiac specific and is not preferred over modern high-sensitivity troponin.
Amylase / lipase	Pancreatic injury; amylase is also salivary	Amylase can enter urine. Lipase is more useful/specific in many current pancreatitis pathways; choose amylase only when it is the best supplied option.
Acid phosphatase (ACP)	Prostatic and other isoenzymes; increased bone turnover may contribute	Older broad ACP questions are not modern definitive prostate or liver diagnostic rules. Do not confuse ACP with ALP.

An AST/LDH/CK-MB pattern is a historical cardiac panel. Pulmonary injury and hemolysis can raise LDH without the same cardiac pattern, but a panel alone does not diagnose pulmonary embolism. BNP/NT-proBNP indicate cardiac wall stress and support heart-failure assessment; D-dimer addresses fibrin turnover in an appropriate thromboembolism assessment. They are not interchangeable. Streptokinase promotes plasminogen activation and hence fibrinolysis; it does not directly hydrolyze fibrin as a normal human digestive enzyme.

Enzyme tests measure activity under defined conditions

Hemolysis, sampling time, tissue distribution and assay method can change interpretation. A high enzyme concentration is not necessarily proof of injury to the organ named in an MCQ. The word pancreatic-specific is itself too strong for total amylase, which has salivary sources.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition - PDF pp. 128-131

Enzyme Kinetics and regulation session 2.pdf - PDF pp. 29-41; enzyme regulation and isoenzymes

Cardiac Biomarkers - Accessed 23 September 2026; supplemental clinical verification - <https://www.ncbi.nlm.nih.gov/books/NBK545216/>

Analytical and Clinical Perspectives on Amylase - Accessed 23 September 2026; supplemental clinical verification - <https://www.ncbi.nlm.nih.gov/books/NBK557738/>

Biochemistry, Lactate Dehydrogenase - Accessed 23 September 2026; supplemental clinical verification - <https://www.ncbi.nlm.nih.gov/books/NBK557536/>

Liver injury, cholestasis and synthetic-function patterns

SOURCE BASIS: past-paper gap audit + book/reference extension

Pattern	Evidence	Reasoning
Hepatocellular injury	ALT/AST predominate	ALT is more liver-enriched, not liver-exclusive; AST also occurs in heart, muscle and RBCs.
Cholestatic pattern	ALP with GGT or 5-prime-nucleotidase	Supports hepatobiliary origin; obstruction, PBC and PSC are examples, not a diagnosis from one number.
Bone-related ALP	ALP elevated with no parallel GGT rise	Consider skeletal growth, Paget disease or other bone turnover; interpret age and pregnancy.
Synthetic impairment	Low albumin and prolonged PT/INR	Albumin changes slowly and has nonhepatic causes; vitamin K deficiency also prolongs PT.
Alcohol-associated pattern	AST often exceeds ALT; GGT may rise	A typical pattern, not an absolute ratio or a universal 300 U/L ceiling.
Very high aminotransferases	Acute viral, toxic or ischemic injury	ALT can exceed 1000 U/L in acute hepatitis; magnitude alone does not identify a virus.

Total circulating AST has a shorter approximate half-life (about 17-18 hours) than ALT (about 47 hours), so ALT can remain elevated longer after an acute episode; ongoing release and mitochondrial AST complicate the pattern. ALT/AST are PLP-dependent, so B6 deficiency and whether reagent supplies PLP affect measured activity. GGT is membrane associated but is not thereby a GPI-anchored protein; ALP is a classic GPI-anchored enzyme.

Dark urine with conjugated bilirubin, pale stools and a cholestatic enzyme pattern indicate impaired bile delivery. Concurrent upper abdominal pain and pancreatic enzyme elevation can suggest biliary pancreatitis, but alcohol history and overlapping patterns prevent certainty without further evaluation. Cystic fibrosis can also cause cholestatic liver disease; an unqualified CF EXCEPT question excluding cholestasis is defective.

Function is not synonymous with enzyme leakage

Cirrhosis can lower albumin and prolong coagulation times while bilirubin rises; none of these is a universal exception. Low aminotransferases in advanced disease do not guarantee preserved hepatic function.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition - PDF pp. 128-131; 515-520

Liver Function Tests - Accessed 23 September 2026; supplemental clinical verification - <https://www.ncbi.nlm.nih.gov/books/NBK482489/>

ALT Test: half-life and assay interpretation - Accessed 23 September 2026; supplemental clinical verification - <https://www.ncbi.nlm.nih.gov/books/NBK559278/>

Clinical Methods: Alkaline Phosphatase and GGT - Accessed 23 September 2026; supplemental clinical verification - <https://www.ncbi.nlm.nih.gov/books/NBK203/>

Renal biochemistry: urea, creatinine, albumin and tubular function

SOURCE BASIS: past-paper gap audit + book/reference extension

Urea contributes most of the usual serum nonprotein nitrogen pool; creatinine, urate, amino acids and ammonia also contain nonprotein nitrogen. Urea production depends on hepatic synthesis and protein catabolism. Creatinine originates from the creatine pool and depends on muscle mass as well as renal removal. Azotemia is the biochemical accumulation of nitrogenous waste; uremia is a clinical syndrome, not its exact synonym.

Test / state	What it asks	Limit
Creatinine-based eGFR	Approximate glomerular filtration from an endogenous marker	Muscle mass, secretion and assay interference matter; normal creatinine does not exclude early disease.
Clearance = $\frac{\text{urine concentration} \times \text{flow}}{\text{plasma concentration}}$	A volume of plasma cleared per time	Creatinine clearance modestly overestimates filtration because creatinine is also secreted.
Urine albumin/creatinine ratio	Early evidence of glomerular albumin leakage	A spot ratio reduces dilution effects; persistent abnormality needs confirmation.
Urine/plasma osmolality	Tubular concentrating performance	Depends on hydration, ADH and medullary gradient; not a uniquely diagnostic injury test.
Prerenal volume depletion	Urea can rise disproportionately to creatinine	Vomiting/diarrhea lower perfusion and increase urea reabsorption.
GFR below 15 mL/min/1.73 m ²	G5 kidney failure range if chronic	A single acute result does not establish chronic ESRD or a dialysis decision.

Creatinine assays include Jaffe and enzymatic methods. Hemoglobin, bilirubin, ketones and other interferents have method-specific effects. A source answer saying high hemoglobin causes falsely

low creatinine is not a universal laboratory rule: consult the reagent interference specification. Do not substitute a corrected guessed result for a repeated validated assay.

Triamterene blocks epithelial sodium channels in principal cells, reducing sodium entry and potassium secretion, so it is potassium sparing and may cause hyperkalemia. It is not an aldosterone-receptor antagonist. Nephrogenic diabetes insipidus is resistance to vasopressin, including V2-receptor or aquaporin-2 defects; dilute polyuria is a water-handling problem, different from glucose-driven osmotic diuresis.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition - PDF p. 525

Guyton and Hall Textbook of Medical Physiology (local 2026 PDF) - PDF pp. 356-357, 366-367, 379-380, 426

Renal Function Tests - Accessed 23 September 2026; supplemental clinical verification - <https://www.ncbi.nlm.nih.gov/books/NBK507821/>

Clinical Methods: BUN and Creatinine - Accessed 23 September 2026; supplemental clinical verification - <https://www.ncbi.nlm.nih.gov/books/NBK305/>

Hormone receptors, second messengers and steroid transport

SOURCE BASIS: past-paper gap audit + book/reference extension

Hormone / mechanism	Signal	Consequences
Insulin	Receptor tyrosine kinase	Metabolic transport/storage and protein signaling; not a Gs-cAMP receptor.
Glucagon; vasopressin V2	Gs -> adenylyl cyclase -> cAMP -> PKA	Hepatic fuel mobilization / collecting-duct aquaporin-2 recruitment.
Gq-linked hormones	Phospholipase C -> PIP2 cleavage into IP3 and DAG	IP3 releases stored calcium; DAG activates PKC.
Steroid hormones	Intracellular/nuclear receptors	Cholesterol-derived hormones regulate transcription; their effects are not restricted to this one mechanism.
Thyroid hormones	Nuclear receptors	Tyrosine-derived amines with lipid-soluble behavior; do not assume every amine uses a surface receptor.
NO	Soluble guanylyl cyclase -> cGMP	Smooth-muscle relaxation; distinct from cAMP.

SHBG binds circulating testosterone and estradiol; albumin also contributes transport. Changes in binding protein can change total hormone without the same change in free biologically available hormone. Steroids are largely synthesized on demand rather than stored as mature hormones in secretory granules. Peptide hormones such as insulin are stored in granules. Thyroid hormones are an important exception: they are stored extracellularly in follicular colloid while still part of thyroglobulin.

5-alpha-reductase generates DHT from testosterone in androgen-responsive tissues; deficient activity impairs DHT-dependent differentiation and action. It is not the enzyme for cortisol production

or estrogen aromatization. In nephrogenic diabetes insipidus, V2 signaling or AQP2 response fails despite available ADH; central diabetes insipidus instead lacks appropriate ADH secretion.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition - PDF pp. 433-447; 562-580

Guyton and Hall Textbook of Medical Physiology (local 2026 PDF) - PDF pp. 379-380; chapter 75, hormone chemistry and receptors; chapter 81, androgen conversion

Thyroid biochemistry and metabolic laboratory patterns

SOURCE BASIS: past-paper gap audit + book/reference extension

Follicular cells synthesize thyroglobulin, transport iodide into the follicular lumen and use thyroid peroxidase to iodinate its tyrosines. MIT and DIT coupling forms T3 and T4 within the protein. Endocytosis and proteolysis of thyroglobulin release hormones; iodide is recycled from uncoupled iodotyrosines. This differs from storing free peptide hormone inside secretory granules.

State	Typical TSH / free T4	Metabolic interpretation
Overt primary hypothyroidism, including established Hashimoto disease	High TSH / low free T4	Reduced thyroid output removes negative feedback. LDL commonly rises through impaired clearance.
Overt primary hyperthyroidism	Suppressed TSH / raised free T4 and/or T3	Increased metabolism and glucose turnover; glucose intolerance may occur, LDL commonly falls.
Subclinical or central disorders	Not the same paired pattern	Hashimoto disease can have other phases; low T4 with low/inappropriately normal TSH suggests central dysfunction.

A high-glucose/low-LDL answer in a hyperthyroid exam vignette describes a tendency, not mandatory hyperglycemia in every patient. Likewise HDL does not consistently decrease in hypothyroidism; this makes simplistic triglyceride/HDL EXCEPT questions context dependent.

Sources for this section

Guyton and Hall Textbook of Medical Physiology (local 2026 PDF) - PDF pp. 929-940; thyroid synthesis, metabolic effects and feedback

Endotext: Thyroid Hormone Synthesis and Secretion - Accessed 23 September 2026; supplemental clinical verification - <https://www.ncbi.nlm.nih.gov/books/NBK285550/>

Past-paper audit: flawed items and conventions to keep separate

SOURCE BASIS: past-paper gap audit + book/reference extension

The available papers test both established metabolism and older clinical-laboratory conventions. Review the mechanism first, then identify exactly what the question asks and which choices were supplied. The app retains original source numbering and differentiates editorial answers from source marks. Twenty-three source items remain ungraded because the choices, wording or evidence do not support one reliable key.

Source problem	Learn instead
No UDP choice for cellulose; reversed succinyl-CoA reaction; no three-disulfide answer for insulin	Cellulose uses UDP-glucose; succinyl-CoA -> succinate yields GTP; insulin has three bridges in total.
All listed citrate effects or liver reactions can be true	Citrate inhibits PFK-1, activates ACC and exports acetyl carbon; liver can perform the listed synthesis/oxidation pathways.
CF cholestasis excluded; excess hepatic cholesterol assumed to diagnose a disease	CF can cause cholestasis. Cholesterol imbalance establishes accumulation, not a unique diagnosis.
Azotemia equated to uremia; all listed NPN compounds actually contain nitrogen	Separate biochemical accumulation from clinical syndrome; classify compounds by nitrogen outside proteins.
Universal Gilbert urobilinogen change; all-true neonatal jaundice or cirrhosis EXCEPT lists	Use conjugated/unconjugated fractions and mechanism; do not memorize nonexistent universal exceptions.
Unverified SLOS/mevalonic hematology or sotolon differential claims	Use established pathway blocks and disease markers; defer unsupported discriminator claims.
MSUD choice missing leucine/ketoacids; brain excluded from creatine synthesis	Teach the actual toxic metabolites and acknowledge extrahepatic creatine synthesis.
Ambiguous cataract/AGE, BNP/D-dimer or GPI-anchor wording	Separate polyol from glycation, wall stress from thrombosis, and membrane association from GPI anchoring.

Coverage is not an exam prediction

All source occurrences have a recorded destination in the coverage audit. That includes principles explaining why an item is flawed. It does not turn 23 unresolved questions into scored questions or certify missing lecture content. The separate reconstruction remains an optional supplement.

Sources for this section

Lippincott Illustrated Reviews: Biochemistry, Ferrier, 6th edition - PDF pp. 143-155; 246-265; 410-414; 507-555; 562-582

Coverage limits and next sources

- The full 103-page carbohydrate slide PDF and 42-page first lipid slide PDF were located in BIOCHEM/05 Metabolism during the September 2026 audit; the earlier claim that the carbohydrate deck was unavailable is superseded.
- Later Term 2 lipid, nitrogen, nucleotide, endocrine and clinical-diagnostic lecture decks have not been established in the supplied files. Relevant past-paper topics are now taught using the inspected books and explicitly cited supplementary clinical references, not falsely attributed to a missing lecture.
- The 15 collections contain 641 question occurrences, not 641 independent concepts. One collection is an unauthenticated reconstruction. Twenty-three flawed or unresolved items remain ungraded; their ambiguity must not be converted into a false biological rule.
- The local Ferrier file is 6th edition, whereas the lipid lecturer refers to 7th edition. Modern and historical ATP accounting and cardiac-marker conventions are distinguished.
- Coverage has been checked against available source questions and the available lecture/book topics. Missing lecture files and future examination content cannot be certified complete. Read original images and qualifiers alongside the synthesis.

Lecture transcript archive

Full original files are embedded in this PDF. Open the PDF attachments panel in a reader that supports attachments, or use `_build/transcripts` in this folder. These are uncorrected source records, not an answer key.

Biochemistry Metabolism Recorded Lecture - 1h04m20s.transcript.txt

IMG_4406.transcript.txt

IMG_4407.transcript.txt

Lipid metabolism Esmaeili 1.1.transcript.txt

Lipid metabolism Esmaeili 1.2.transcript.txt

Lipid metabolism1 Esmaeili 1.3.transcript.txt

Local source inventory

Exact paths and source hashes are stored in `_build/source-manifest.json`. Section references distinguish original evidence from book additions.

Biochemistry [Lippincott's Illustrated Reviews Series] Denise R. Ferrier PhD.pdf

Guyton and Hall Textbook of Medical Physiology (2026).pdf

Enzyme Kinetics and regulation session 2.pdf

Lipid metabolism 1.pdf

carbohydrate metabolism MD Kish.pdf

Biochemistry Metabolism (Part 1) Notes - Hawraa.pdf

Biochemistry metabolism notes (pt 2) slides 27-80 .pdf

Biochemistry Metabolism Recorded Lecture - 1h04m20s.transcript.txt

IMG_4406.transcript.txt

IMG_4407.transcript.txt

Lipid metabolism Esmaeili 1.1.transcript.txt

Lipid metabolism Esmaeili 1.2.transcript.txt

Lipid metabolism1 Esmaeili 1.3.transcript.txt